

COMP90024 Assignment 2

Team 71

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GitLab repository address: <https://gitlab.unimelb.edu.au/ANLIA0/comp90024-a2-t71>

Youtube demonstration video address: <https://youtu.be/eqvnE3FC1Uw>

Abstract

This project presents a cloud-native big data analytics system designed to integrate social media content with government datasets to investigate key societal issues in Australia. Leveraging Kubernetes, Fission, and Elasticsearch on the Melbourne Research Cloud, the system enables scalable data ingestion, real-time querying, and interactive visualisation through Jupyter Notebook. The platform supports diverse scenario-based analyses, including traffic congestion, housing sentiment, inflation trends, and political discourse. By applying techniques such as sentiment analysis and correlation analysis, we uncover insights into the relationship between online engagement and real-world conditions. Findings reveal, among others, a moderate alignment between Reddit engagement and inflation, and a negative correlation between rising house prices and Mastodon sentiment. While data sparsity and API limitations posed challenges, the system demonstrates the feasibility of using cloud infrastructure to support timely, data-driven social analysis.

1 Introduction

1.1 Motivation and objectives

The exponential growth of user-generated content on social network platforms presents a valuable opportunity to analyse public comments at scale. With the increasing availability of cloud computing infrastructure, scalable big data analytics systems have become indispensable to revealing trends, opinions, and behaviours across diverse online communities. This project aims to harness these technological advancements to investigate issues of public interest in Australia by combining social media data with structured government datasets.

Our team developed a cloud-native solution hosted on the Melbourne Research Cloud (MRC). The system employs Kubernetes for coordination, Fission for event-driven data ingestion, and

ElasticSearch (ES) for real-time data storage and retrieval. The analytical front-end is implemented in Jupyter Notebook, offering an interactive interface for data visualisation and scenario-based exploration.

To ensure relevance to the Australian context, we developed heuristics to infer geographic associations from data lacking explicit location metadata. These include identifying Australia-related subreddits, Mastodon server domains, and content-specific keywords. By integrating unstructured social media data from platforms with structured datasets from official government and open data platforms, our system supports a broad range of scenario-driven analyses. The result is a flexible, cloud-based platform for uncovering trends in public sentiment, behavioural patterns, and policy impact across Australian society.

1.2 Overview of data sources

To support comprehensive and multifaceted analytics, this project utilises a diverse collection of datasets drawn from both social media platforms and authoritative government sources. These datasets span real-time streams and historical records, enabling both longitudinal analysis and short-term trend detection.

Unstructured content and metadata were harvested from Mastodon, Reddit, YouTube, and Last FM. The Mastodon data includes a large-scale historical archive (mastodon-144g) covering the period from 2010 to 2025, obtained from the SPARTAN repository. Reddit data consists of submissions and comments collected using the Reddit Application Programming Interface (API), filtered to capture Australian-relevant discussions such as elections, inflation, and housing. YouTube metadata was obtained for videos focused on Melbourne lifestyle and Australian political events, while Last FM data includes weekly top track charts and listener counts used to study behavioural patterns.

Structured datasets from trusted Australian institutions were also incorporated. Crime statistics and dwelling data, including transfer counts and median property prices, were sourced from the ABS. Inflation and Consumer Price Index (CPI) data were obtained from the RBA. Number of hospitalised injuries from road crashes was sourced from Department of Infrastructure, Transport, Regional Development, Communications and the Arts (DITRDCA). Additionally, traffic volume and congestion data were acquired from Bluetooth sensor feeds via the DataVic platform, with a collection frequency of every two hours.

The integration of unstructured social media content with structured government data enables a comprehensive analytical framework that supports a wide range of investigations into Australian public life. All datasets are ingested and indexed within the ES database to support efficient querying and interactive visualisation through the Jupyter Notebook interface. A summary of all datasets used in this project is presented in Table 1.

1.3 Scenarios and their connection to Australia

Our project explores a diverse set of analytical scenarios centred on Australian societal, political, and economic contexts. These scenarios integrate unstructured social media discourse and structured government datasets to investigate real-world trends and behaviours.

Dataset	Source	Description
mastodon-144g	SPARTAN [1]	Historical posts from 2010–2025
reddit	Reddit API [2]	Posts and comments on selected topics
youtube	YouTube API [3]	Metadata on lifestyle/election-related videos
crime data	ABS [4, 5]	Victim and offender statistics by category
CPI data	RBA [6]	Consumer Price Index and inflation series
traffic volume	Bluetooth API - DataVIC [7]	Road density and congestion (2-hour intervals)
traffic injuries	DITRDCA [8]	Number of hospitalised injuries from road crashes
music listeners	Last FM API [9]	Top-50 tracks and listener counts (daily)
dwelling data	ABS [10]	Median price and number of house transfers

Table 1: Summary of Data Sources

1.3.1 Urban Mobility and Societal Behaviour

We analysed real-time traffic data in Victoria to assess congestion patterns across time and geography. This includes visualising traffic density over the past two hours, identifying peak congestion periods, and evaluating injury statistics by location. We further investigated the interplay between traffic activity and cultural behaviour by examining whether traffic intensity correlates with music-listening patterns using Last FM data.

1.3.2 Public Sentiment on Crime and Housing

We conducted temporal sentiment analysis using Mastodon posts filtered by crime- and housing-related keywords. These sentiments were compared against official crime victimisation and offender rates, as well as housing price and transfer trends, to assess the alignment between public perception and real-world statistics. Regression analysis revealed a moderate inverse relationship between rising housing prices and sentiment after November 2022, reflecting growing public concern over affordability.

1.3.3 Reddit and YouTube Analysis on National Events

To understand public discourse around the Australian federal elections, we performed hot-word and sentiment analysis on Reddit posts, comparing pre- and post-election periods. Additionally, we analysed YouTube video metadata related to Melbourne lifestyle, computing engagement scores and exploring their lagged correlation with traffic congestion. A modest positive correlation was found between video popularity and future congestion, hinting at possible behavioural links.

1.3.4 Inflation and its Broader Implications

We compared Reddit voting patterns on inflation-related posts with the official Consumer Price Index (CPI), uncovering a moderate positive correlation. Topic modelling further revealed key themes such as cost of living, rent pressure, and energy prices. We also explored the relationship between inflation and crime, finding offence-specific patterns—such as strong correlation between CPI and blackmail/extortion—that challenge the blanket notion of “more inflation, more crime.”

2 System Architecture

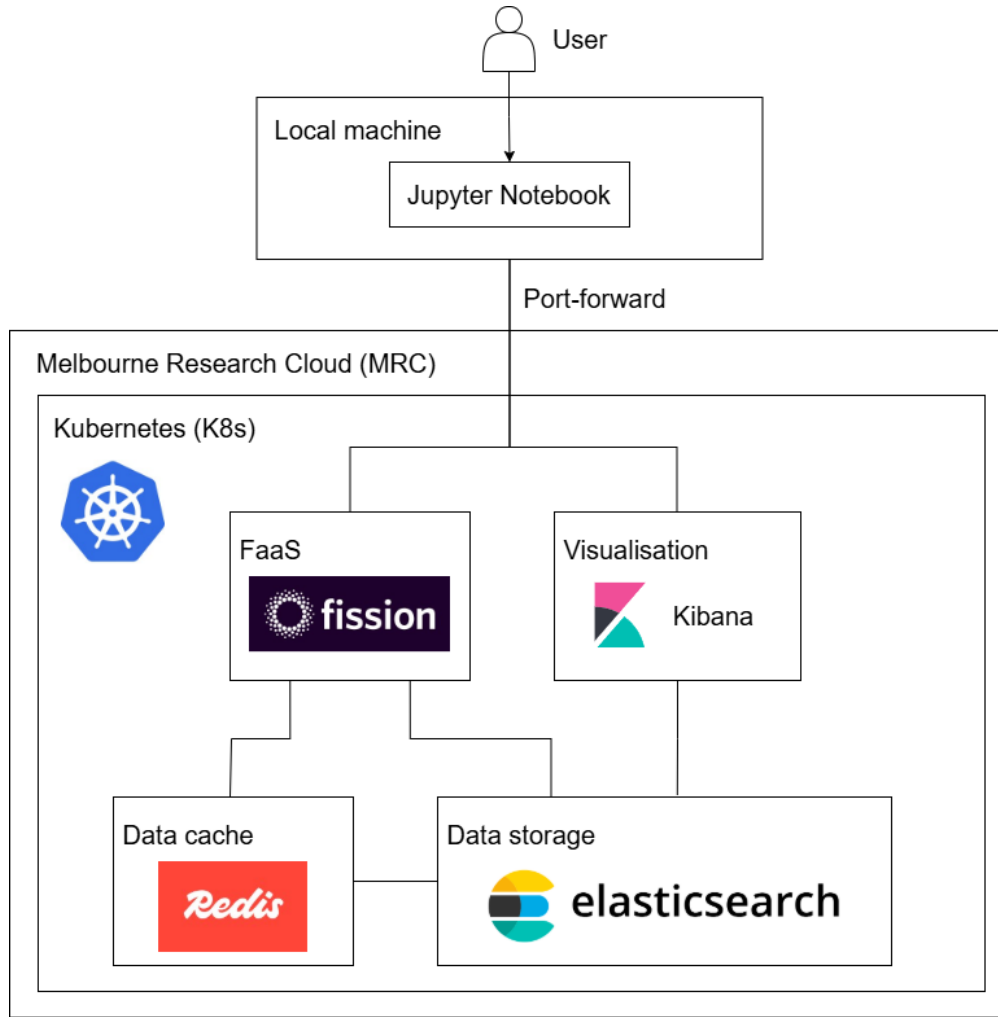


Figure 1: Overall system architecture

2.1 Infrastructure Design

2.1.1 Kubernetes

Our Kubernetes (K8s) cluster is deployed on the MRC, consisting of one master node and three worker nodes. The master node functions as the control plane of the cluster and is responsible for essential coordination tasks. It manages the API server, which facilitates communication between user commands and the cluster, and maintains the overall configuration and state of the system using `etcd`. The master node also includes the `kube-scheduler`, which assigns newly created Pods to appropriate worker nodes, and various controllers that ensure the actual state of the cluster matches the desired state. The overall structure of this setup is illustrated in Figure 2.

The three worker nodes serve as the computational backbone of the cluster, running application workloads such as Fission functions and Elasticsearch services. These nodes support scalability and resilience by distributing Pods across the infrastructure. The `kube-scheduler` places new

Pods based on available resources, promoting even distribution. Controllers such as ReplicaSets automatically redeploy the affected Pods on healthy nodes if a node becomes unavailable, preserving service continuity.

Overall, this Kubernetes deployment provides a robust, fault-tolerant, and scalable environment that supports efficient resource utilisation and maintains high availability of services.

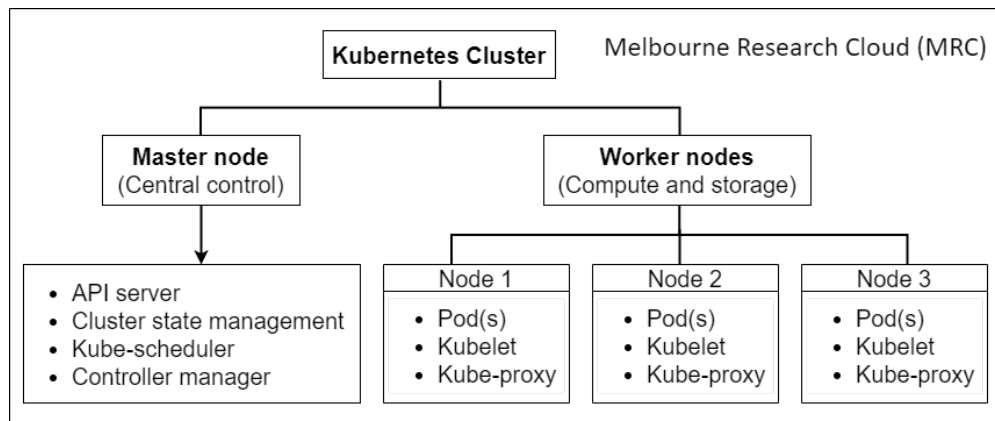


Figure 2: Structure of Kubernetes

2.1.2 Network

In our Kubernetes cluster, we use a ClusterIP Service Network, which provides secure and efficient internal communication between Pods and services within the cluster. Each service is assigned a unique and stable ClusterIP address, allowing multiple Pods to be accessed under a single virtual IP. This internal IP is only accessible within the cluster, ensuring a secure communication.

Pods communicate with services through this ClusterIP network, which automatically distributes traffic across all healthy Pods that match the service selector, achieving load balancing without any external exposure. This setup is ideal for internal microservices communication, where services like Elasticsearch, Redis, and Fission functions can efficiently exchange data without external access.

To make certain services accessible externally, we utilise port-forwarding, a method that securely exposes specific service ports on local machines. Forward the port allows external clients (Jupyter Notebook on a local system) to interact with internal services like representational state transfer (ReST) APIs and Elasticsearch. This approach provides a secure and controlled method of external access without exposing services directly to the public network.

While port-forwarding is convenient, it is not recommended for use in a production environment. It is temporary and lack of high availability, users can only use in their own local machines. Moreover, port-forwarding does not inherently provide secure communication via HTTPS (Hypertext Transfer Protocol Secure), as it lacks built-in support for TLS/SSL (Transport Layer Security / Secure Sockets Layer) encryption.

Any data sent through port-forwarding is transmitted in plain text, which is a security risk for sensitive data. We explored more secure alternatives, such as using Nginx as an ingress controller. However, our cloud environment does not support LoadBalancer services, which prevents us from using Nginx effectively. As a workaround, we have considered using NodePort to expose specific ports for external access. Due to time constraints, we continue using port-forwarding for external

access, while acknowledging its limitations.

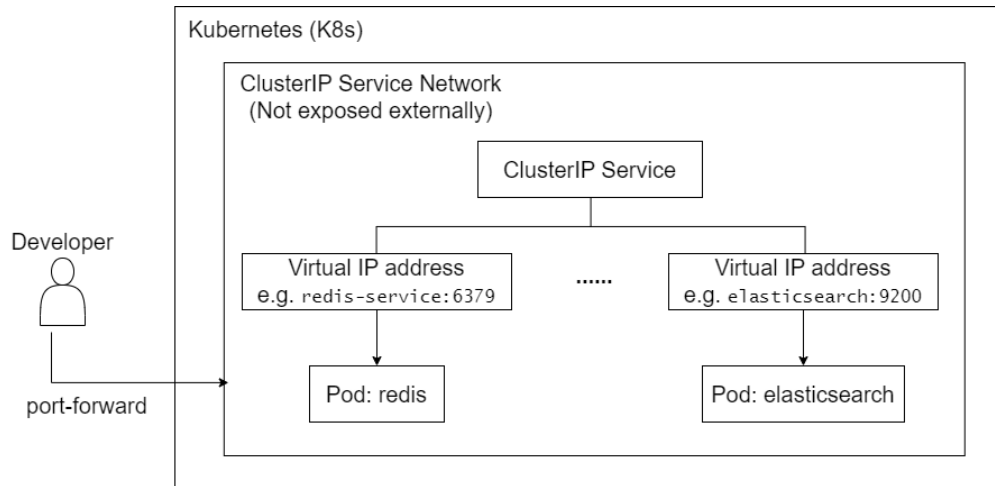


Figure 3: An example of usage for ClusterIP Service Network

2.1.3 Storage

In our Kubernetes cluster, persistent storage is provided through OpenStack Cinder Block Storage using a custom `perfretain StorageClass`. This class is configured with a `Retain` reclaim policy, ensuring that the underlying PersistentVolumes (PVs) are preserved even if their associated PersistentVolumeClaims (PVCs) are deleted.

We use this storage class to provision two 100 GiB PVCs for Elasticsearch, supporting large-scale indexing and search operations, as well as an 8 GiB PVC for Fission to persist configuration and function state.

This setup offers a robust and efficient storage solution, combining the performance of Cinder Block Storage with reliable data retention, ensuring critical application data remains available across pod restarts and cluster reconfigurations.

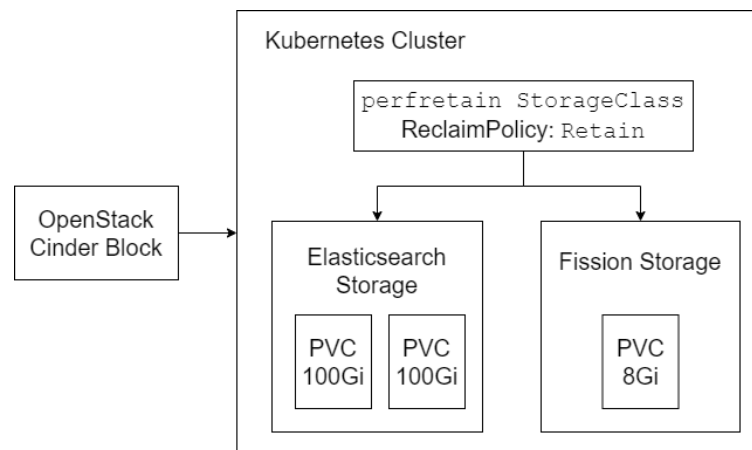


Figure 4: Storage configuration

2.2 Service in Kubernetes

2.2.1 Fission

In our Kubernetes cluster, Fission is used as a Function-as-a-Service (FaaS) platform for deploying data harvester functions and ReST API services. This setup enables the rapid deployment and execution of serverless functions, ensuring efficient resource utilisation. With this setup, we have deployed multiple HTTP-triggered and timer-based Fission data harvester functions. These functions operate on a serverless model, meaning they are only active when triggered, minimising resource consumption. Once triggered, these harvesters collect data, process it within the function, and directly store the processed data in Elasticsearch for further analysis.

In addition to data harvesting, we also host a set of ReST APIs within Fission. These APIs provide a secure and efficient interface for accessing data in Elasticsearch. The APIs are exposed within the cluster using a ClusterIP Service, ensuring secure internal communication. External access to these APIs is managed through port-forwarding, allowing secure connections from the frontend.

Fission's serverless architecture ensures that Pods are allocated only when a function is actively running, providing an efficient and cost-effective solution for both data harvesting and API management. This dynamic allocation minimises resource consumption and allows for seamless scalability. With this architecture, new harvester functions or APIs can be quickly added, updated, or removed without impacting the overall cluster, ensuring a flexible and efficient system.

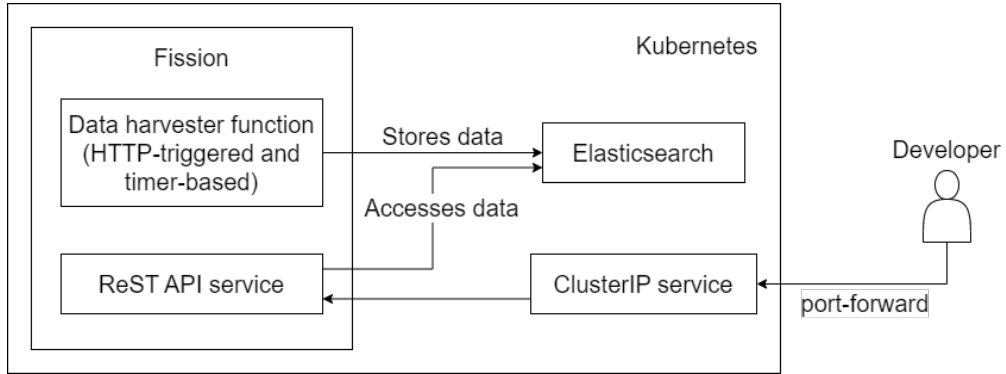


Figure 5: Fission-based serverless data pipeline in Kubernetes

2.2.2 ElasticSearch

In our Kubernetes cluster, Elasticsearch is deployed as a scalable and reliable engine for data storage, indexing, and search. It enables rapid ingestion of large datasets, real-time querying, and advanced text analytics. We use a dual-master setup with two nodes, `elasticsearch-master-0` and `elasticsearch-master-1`, each with 100 GiB of storage. These master nodes also serve as data nodes for managing indexes.

Each harvester writes to a dedicated index, ensuring category-level data isolation (e.g., election or lifestyle topics). Elasticsearch automatically overwrites documents with duplicate IDs, maintaining a clean and up-to-date dataset. This is particularly useful for incremental data collection, allowing metadata updates (e.g., view counts, titles) without redundancy. Bulk operations and flexible schemas further support scalability and consistency.

Custom index mappings are defined to control data types and search behavior, reducing type ambiguity and improving query reliability. Fields such as timestamps, numerical metrics, and text are explicitly typed to support accurate filtering, aggregation, and full-text search.

Elasticsearch also supports dynamic mapping expansion. As harvesters extract new metadata, previously unseen fields are automatically added to the schema with inferred types, allowing the data model to evolve without manual intervention.

2.2.3 Kibana

In our Kubernetes project, Kibana is deployed as a data visualisation and analytics platform, offering an intuitive web interface for exploring and analysing data stored in Elasticsearch.

Kibana enables us to create dynamic data views, visualise data from various Elasticsearch indices, and perform comprehensive index management. This includes inspecting index attributes, monitoring data health, and managing index settings, providing a user-friendly way to interact with our data.

2.2.4 Redis

In our Kubernetes project, Redis is deployed as a high-speed, in-memory data store and cache, ensuring fast data access and efficient caching for our API services. Redis is configured as an in-memory cache, allowing frequently accessed data to be temporarily stored. In our ReST API services, results from processed API requests can be cached in Redis. This setup ensures that subsequent requests for the same data are served instantly from the cache, eliminating the need to reprocess data in Elasticsearch and significantly improving response times.

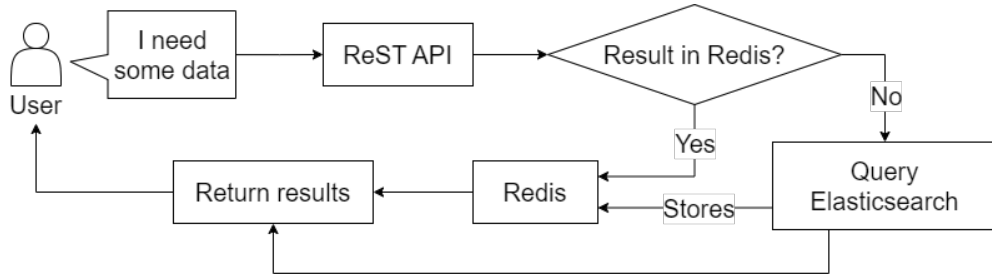


Figure 6: ReST API caching workflow with Redis and Elasticsearch

2.2.5 Jupyter Notebook

We use Jupyter Notebook as a local frontend for our project. Jupyter Notebook offers an interactive coding environment with support for creating visually appealing data visualisations using libraries such as Matplotlib, Seaborn, and Plotly.

Given the nature of our data, which primarily consists of geospatial traffic data and text data, we frequently utilise interactive Mapbox visualisations for geographic data and WordCloud components for text analysis. This setup allows for efficient data exploration and visualisation in a user-friendly interface.

2.3 Development and Operations

2.3.1 Version-Control repository

Our team uses GitLab as our version control platform, leveraging its robust features for secure and efficient code management. We maintain a private team repository with strict access controls to ensure data privacy and secure collaboration.

Each team member works on their own private branch, developing and testing code independently. This approach minimises conflicts and maintains code isolation during development. Code that has not been tested is strictly prohibited from being pushed to the master branch. When a branch is deemed mature and stable, it goes through a review process. Upon approval, it is merged into the master branch, ensuring that only well-tested and reliable code is integrated. In cases where multiple branches involve overlapping code, team members use GitLab’s merge request functionality to resolve conflicts collaboratively before final integration. This branching strategy ensures that our team maintains a clean, stable, and secure codebase, allowing for efficient parallel development without risking code integrity.

2.3.2 Unit Test

To ensure the reliability and correctness of our serverless functions and API services, we have developed a comprehensive suite of unit tests using Python’s `unittest` framework. While the harvester functions rely on the Fission runtime in production, their internal logic is designed to be modular and testable. By mocking external dependencies such as API clients and Elasticsearch connectors, we are able to validate the core functionality of these harvesters independently from the Kubernetes runtime environment.

Each test case is designed to cover key functional aspects, such as input validation, Elasticsearch query construction, response structure, and exception handling. In our design, the test logic is organised into two separate modules: one dedicated to testing all API endpoints, and the other focused on verifying all harvester functions.

For the API test module, we simulate HTTP requests using the `requests` library and assert response status codes, content types, and output schemas to ensure accurate and consistent external data access. For the harvester test module, we provide mock inputs to validate data parsing, deduplication logic, and logging behaviors. External dependencies such as the YouTube API and Elasticsearch client are mocked to allow testing the core logic independently of the Kubernetes runtime environment.

Additionally, file-based logs generated during testing are named using sanitized paths and timestamps to facilitate tracking and debugging. All tests are automatically executed in a CI pipeline before deployment, ensuring that only stable and well-tested components are released into the production cluster.

3 Implementation Details

3.1 Data ingestion pipeline

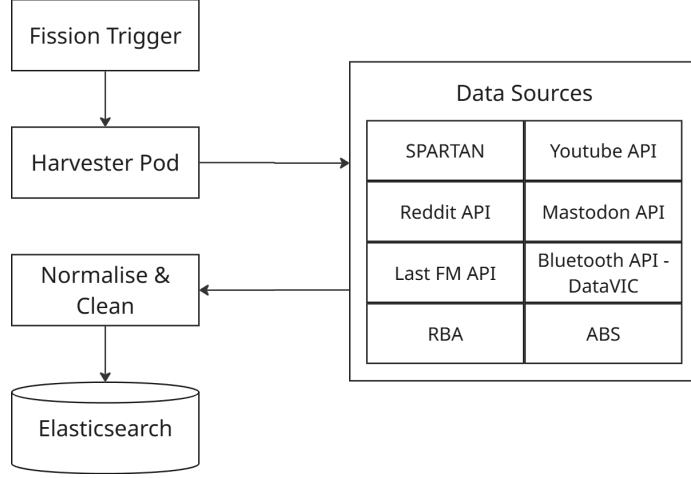


Figure 7: Data ingestion pipeline

The ingestion layer constitutes the first stage of the analytics pipeline. As Figure 7 illustrates, it harvests raw records from a diverse set of external sources—including live Bluetooth traffic probes (DataVic), YouTube uploads, Reddit posts and comments, Mastodon posts, daily Last FM charts, and structured government datasets such as RBA inflation series and ABS crime statistics—and forwards them to Elasticsearch for unified storage and querying.

All harvesters run as Fission functions inside the Kubernetes cluster, so each source is treated as an independent, elastically scalable micro-service. For high-frequency streams (e.g., traffic, YouTube, Reddit, music charts), we register cron-style timers in Fission; when a timer fires, the platform spins up a fresh Pod—or reuses a warm one—to pull the latest slice of data. Lower-frequency but larger static collections (the SPARTAN Mastodon archive, RBA, ABS) are refreshed on demand, preventing idle resource consumption.

Each function applies lightweight preprocessing to standardise the data. Timestamps from different APIs are converted to a common zone and format to support cross-source alignment. Missing or non-finite numeric values (e.g., `NaN`, `Inf`) are mapped to JSON `null` to comply with Elasticsearch’s numeric-field rules. Text fields are stripped of control characters, URLs and platform-specific markup, ensuring downstream NLP components receive clean tokens. No heavyweight language modelling is introduced at this stage: the guiding principle is to preserve semantic richness while enforcing syntactic consistency.

Each cleansed batch is then submitted to Elasticsearch via the `_bulk` endpoint. Bulk ingestion minimises HTTP round-trip overhead and allows Elasticsearch to coalesce disk flushes, thereby smoothing I/O spikes during peak traffic. Every index is provisioned with a replica shard, guaranteeing read availability throughout rolling upgrades or node failures. Once the cluster returns an HTTP 202 acknowledgement, the function terminates, freeing capacity for other tasks.

3.2 Description of APIs, ReSTful services

Table 2: ReST API endpoints provided by the system

Endpoint	URL	Short description
Traffic injury	<code>/trafficinginjury/{gender}/{year}</code>	Injury counts across all categories
Traffic density (hourly)	<code>/trafficagghour/{startdate}</code> <code>/{enddate}</code>	VIC congestion index aggregated by hour
Traffic volume (fine)	<code>/trafficvolumeapi/{startd}</code> <code>/{startt}/{endd}/{endt}</code>	Density, speed and geo-coordinates of probes
Crime victims	<code>/victim/{state}/{method}</code>	Victim numbers or per-100k rates
Crime offenders	<code>/offender/{gender}/{age}/{method}</code>	Offender numbers or per-100k rates
Music top tracks	<code>/musictoptrackapi/{startd}</code> <code>/{startt}/{endd}/{endt}</code>	Top-50 tracks with listener counts
Mastodon keyword sentiment	<code>/mastodonkeywordsentiment/{startd}</code> <code>/{endd}/{kw}</code>	Sentiment time-series for keyword
Mastodon hotwords (live)	<code>/mastodonhotword/{startd}/{startt}</code> <code>/{endd}/{endt}</code>	Top-100 hotwords in recent stream
Mastodon hotwords (archive)	<code>/keywordmastodon/{startdate}/{enddate}</code>	Top-100 words in 144 GB archive
House transfers	<code>/housetransfer/{sy}/{sm}/{ey}/{em}</code>	Median prices and transaction counts
Inflation (RBA)	<code>/inflation/rba?start=&end=</code>	CPI and related series (quarterly)
Inflation posts	<code>/inflation/reddit.posts?start=&end=</code>	Reddit submissions on inflation
Inflation comments	<code>/inflation/reddit.comments?start=&end=</code>	Reddit comments (body, score, parent ID)
YouTube lifestyle	<code>/youtubelifeapi?start=&end=</code>	Lifestyle-related video metadata
YouTube election	<code>/youtubeelectionapi?start=&end=</code>	Election-related video metadata
Aus election comments	<code>/auselectionapi/{startdate}/{enddate}</code>	Reddit comments mentioning the election

The back-end exposes a uniform set of ReST endpoints that decouple the Jupyter Notebook front-end—and any potential third-party client—from the underlying Elasticsearch schema. Every route adopts the pattern `/domain/...`, where `domain` denotes a data family such as traffic, crime, inflation, or youtube. The interface is strictly read-only and relies exclusively on HTTP GET; all filters—temporal windows, geographic scope, and so forth—are encoded either as path segments or explicit query parameters. Because index mappings differ across data families, the contract guarantees only the outer JSON envelope, allowing internal fields to evolve with the schema without breaking existing clients. Switching to a different data family simply involves changing the domain prefix in the URL, promoting modularity and reuse in client-side logic.

3.3 Preprocessing and NLP libraries

When processing Reddit posts and their associated comments, we apply a lightweight natural language processing (NLP) pipeline to uncover latent topics and trace how those topics evolve over time. The text is first normalised: a regular-expression filter removes non-ASCII characters and links, and the remaining content is lower-cased. We then tokenise with NLTK’s `word_tokenize` [11] and lemmatise each token via the WordNet [12] lemmatiser to collapse inflected forms to their base lemmas; tokens shorter than three characters and those found in NLTK’s default English stop-word list are discarded. The surviving tokens are rejoined into space-separated strings, preserving readability while producing input suitable for the subsequent topic model.

Topic extraction is carried out with BERTopic [13], which pairs sentence-level embeddings with

an interpretable TF-IDF (Term Frequency-Inverse Document Frequency) bag-of-words clustering layer. The model runs in English mode with an upper bound of ten topics to balance semantic granularity against training cost. Posts and comments are modelled separately, limiting cross-genre drift and allowing direct comparison between submission headlines and discussion threads. Once training finishes, the resulting topics—together with frequency histograms, similarity heat-maps, and time-series plots—are rendered interactively in the Jupyter Notebook front-end for exploratory analysis.

4 Scenarios and Analytics

4.1 Mastodon Hot Words and Public Sentiment on Crime and House Prices

4.1.1 Identifying Hot Words in Mastodon Posts over Custom Time Intervals

To explore public discourse over time, we implemented a hot-word analysis feature based on the `mastodon-144g` dataset. Within the Jupyter Notebook interface, users are prompted to specify a start and end date to define a time interval of interest. Posts within the selected period are preprocessed to remove noise and standardise text. From this refined corpus, the top 100 most frequent words are extracted, representing the dominant terms or “hot words” during that timeframe. A word cloud is then generated with `wordcloud` [14] to visualise these terms, with word size corresponding to frequency. This interactive visualisation allows users to quickly grasp prevalent themes and topics discussed on Mastodon during the specified period, providing an intuitive overview of public sentiment and trends.

Figure 8 shows an example of a word cloud generated from posts collected between 2023-01-01 and 2023-01-04. As expected, the dominant terms during this period reflect widespread celebration of the new year, with words such as “happy”, “new”, and “year” appearing prominently. This outcome demonstrates the utility of the hot-word analysis in capturing the prevailing themes and public sentiments discussed on Mastodon during a given timeframe.



Figure 8: Hot Words on Mastodon: New Year Discussions (2023-01-01 to 2023-01-03)

4.1.2 Comparative Analysis of Crime Rates and Online Sentiment

Figure 9 presents a comparative view of three key indicators across financial years 2008–2025: the victimisation rate, offender rate, and average Mastodon sentiment related to crime. Sentiment scores are computed as annual averages filtered by crime-related keywords and mapped to corresponding financial years. As discussed in Section 5.3.2, the Mastodon dataset exhibits a sharp increase in post volume after November 2022, resulting in sparse and potentially unrepresentative sentiment data in earlier years. This limitation, combined with the relatively flat sentiment trend in the most recent years, suggests that the observed sentiment values may not meaningfully reflect the underlying variation in crime rates. Consequently, no clear correlation is observed between public sentiment and either the victimisation or offender rates.

The victimisation rate, obtained from the Australian Bureau of Statistics (ABS), is based on selected crime categories including homicide, sexual assault, and kidnapping/abduction. Notably, the recent increase in victimisation rate is largely driven by a significant rise in reported sexual assault cases, which dominate the trend in recent financial years. On the other hand, the offender rate—also from ABS and covering a broader range of crime categories—shows a long-term decline from 2015 onwards. This divergence highlights a possible mismatch between victim reporting patterns and offender identification or apprehension, and may reflect broader societal, legal, or systemic dynamics rather than changes in public sentiment.

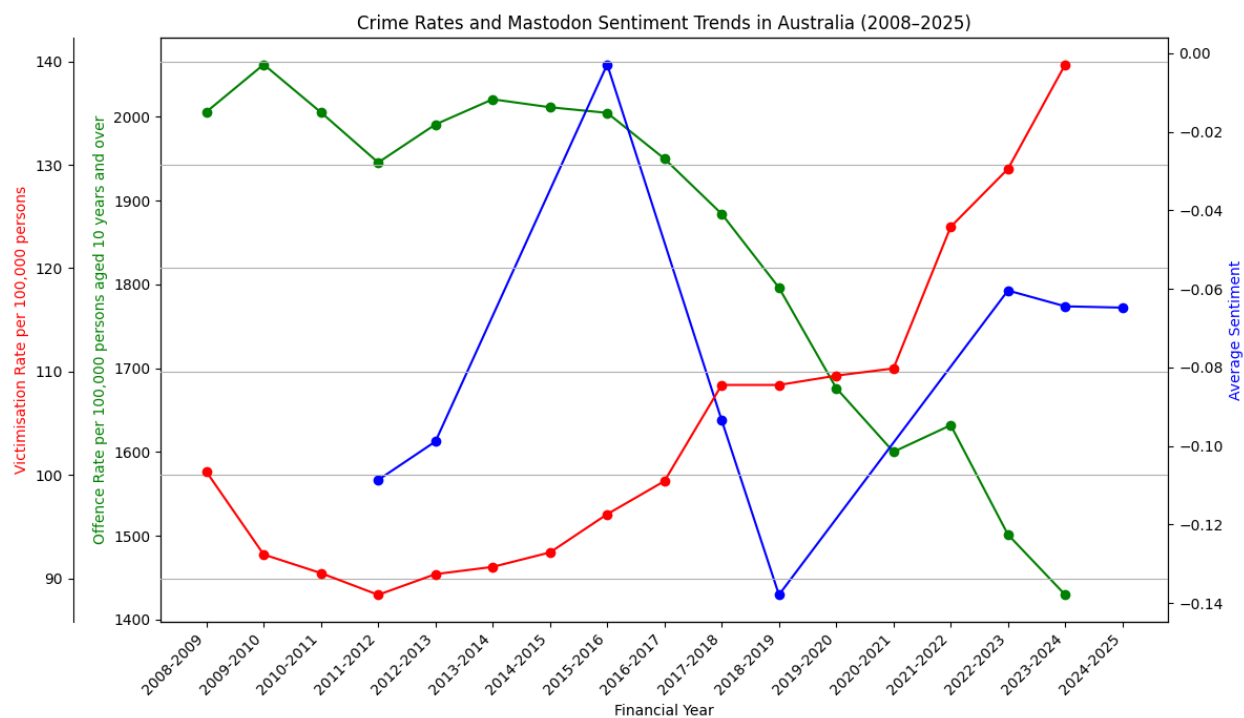


Figure 9: Comparison of victimisation rates, offence rates, and average Mastodon sentiment related to crime from 2008 to 2025.

4.1.3 Public Sentiment and Housing Market Trends in Australia (2010–2024)

Figure 10 presents a three-way temporal comparison between average sentiment on Mastodon regarding housing, the number of dwelling transfers, and the median price of residential properties

in Australia from 2010 to 2024. The sentiment values represent quarterly averages derived from posts filtered using housing-related keywords. In contrast, the median price and transfer volumes were sourced from the Australian Bureau of Statistics (ABS). An obvious inconsistency is visible in the sentiment trend prior to November 2022, due to the low and uneven number of posts in the `mastodon-144g` dataset before the platform experienced a significant usage surge. This data sparsity contributes to the pronounced fluctuations observed in the early sentiment values.

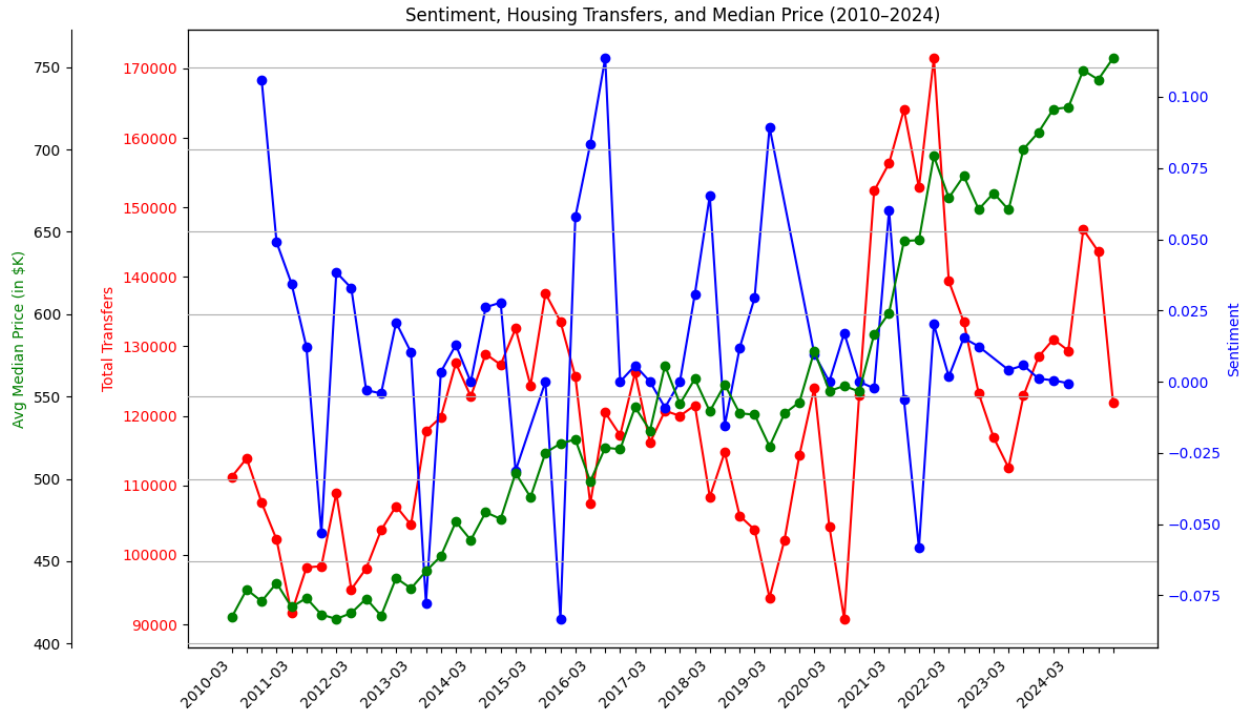


Figure 10: Temporal trends of housing-related Mastodon sentiment (blue), total number of housing transfers (red), and median residential property price (green) from 2010 to 2024.

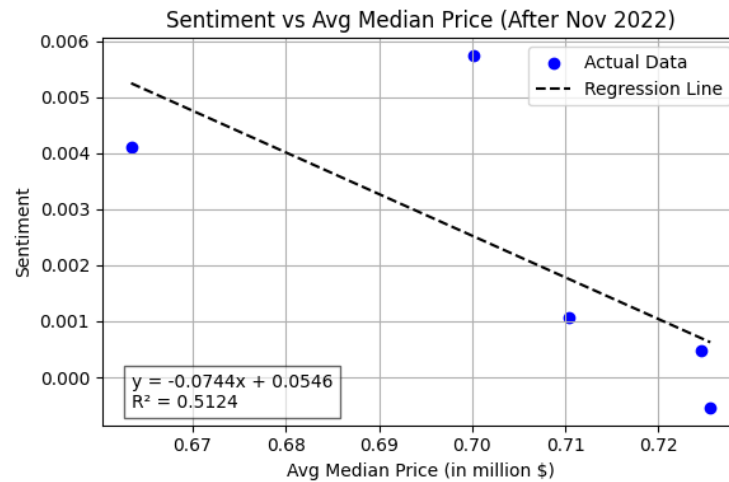


Figure 11: Scatter plot of average Mastodon sentiment versus average median residential property price (in million AUD) after November 2022.

Following the November 2022 surge in Mastodon activity, the sentiment trend stabilises and exhibits a gradual decline, while the median price continues to rise steadily. The number of housing transfers, however, demonstrates periodic fluctuations without a clear long-term trend. To explore the relationship between public sentiment and housing prices post-surge, a linear regression was performed using the data from November 2022 onward (Figure 11). The regression analysis reveals a moderately negative correlation ($R^2 = 0.5124$), suggesting that as median house prices increased, public sentiment became slightly more negative. While this relationship is not definitive, it may reflect growing public frustration or pessimism around housing affordability during this period.

4.2 Traffic in Victoria

In our traffic dashboard, users are empowered to gain latest insights into traffic conditions in Victoria (VIC) with ease. By simply inputting their geographical location in terms of latitude and longitude, users can access up-to-date traffic information tailored to their specified area. Additionally, they can define a specific time period for their query, allowing them to review historical traffic patterns or forecast traffic conditions at a future time. This interactive and user-friendly dashboard ensures that users can make well-informed travel decisions, enhancing their daily commuting experience.

4.2.1 Traffic Mapbox in last 2 hours

Users can view an interactive Mapbox [15] map centred on their specified location. As illustrated in Figure 12, this Mapbox visualisation dynamically displays the cumulative traffic density on each road across VIC over the past two hours. This real-time visualisation provides users with a clear and intuitive understanding of traffic conditions, enabling them to make informed decisions based on the latest data.

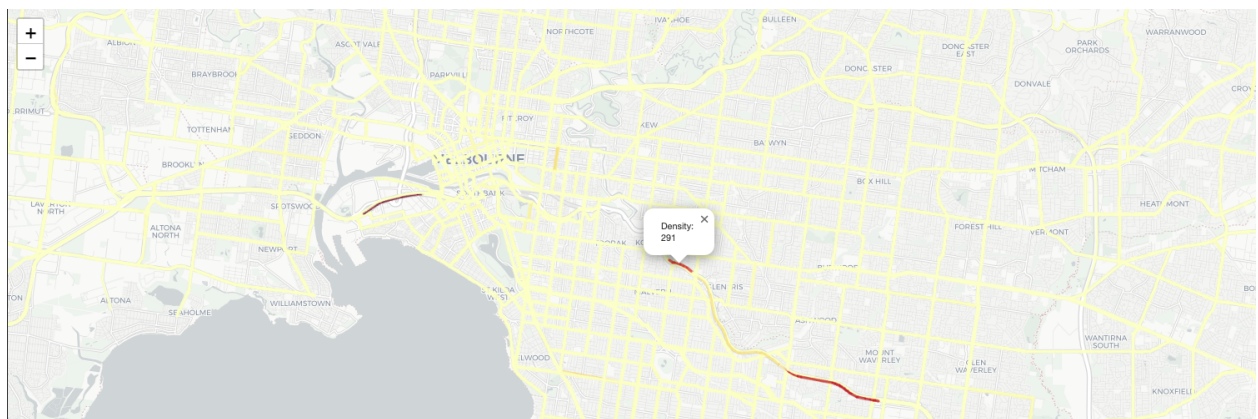


Figure 12: Road density in VIC in the past 2 hours

Users can then explore a detailed heatmap, as shown in Figure 13, which visualises the busiest two-hour intervals of traffic across VIC. The heatmap uses a gradient colour scheme, with deeper red hues indicating higher traffic density during that specific two-hour period. This intuitive representation allows users to identify peak traffic times and make informed travel decisions quickly.

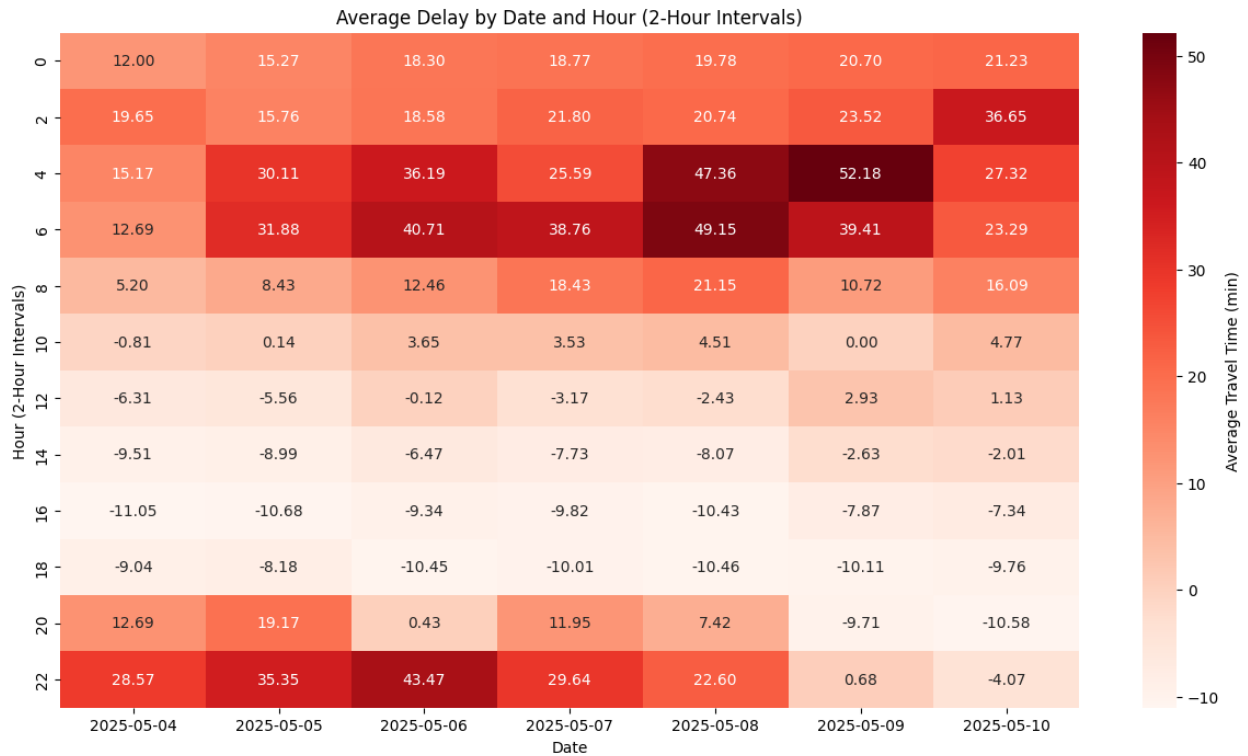


Figure 13: Traffic Heatmap showing average delay in VIC in last 7 days

Our dashboard also offers a comprehensive trend line chart and bar chart, as depicted in Figure 14, which present the average speed and average congestion level across two-hour intervals over the past seven days in Victoria (VIC). For improved visualisation, the congestion level values are scaled by a factor of 100. This dual-chart display allows users to easily observe traffic patterns, track fluctuations in speed, and understand congestion trends throughout the week. The missing data is due to our API harvester reaching its maximum weekly request limit.

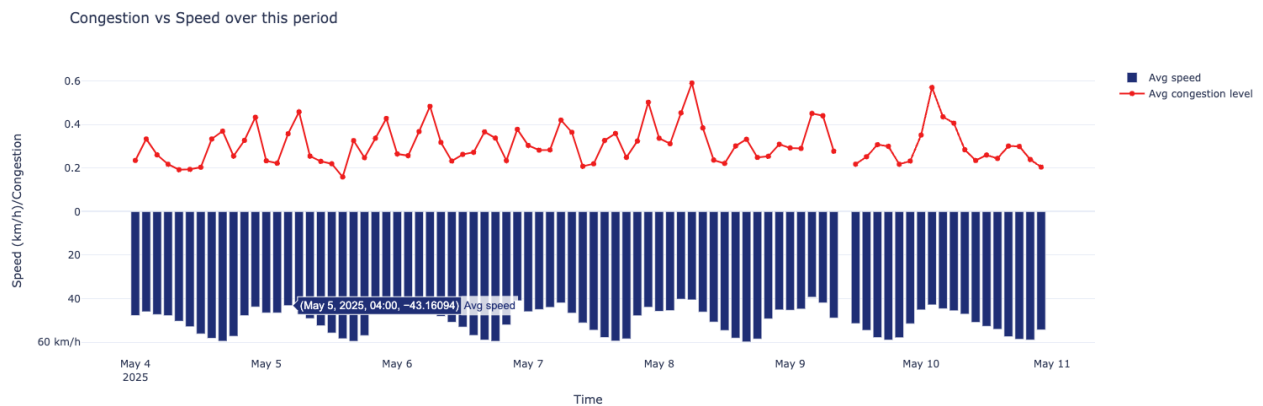


Figure 14: Congestion (Scaled by 100) vs Speed in last week in VIC

Our dashboard also displays the annual traffic injury report to users as shown in Figure 15 and 16. However, due to data limitations, we can only access traffic injury data up to 2021. As a result,

the most recent report available on our dashboard is for the year 2021.

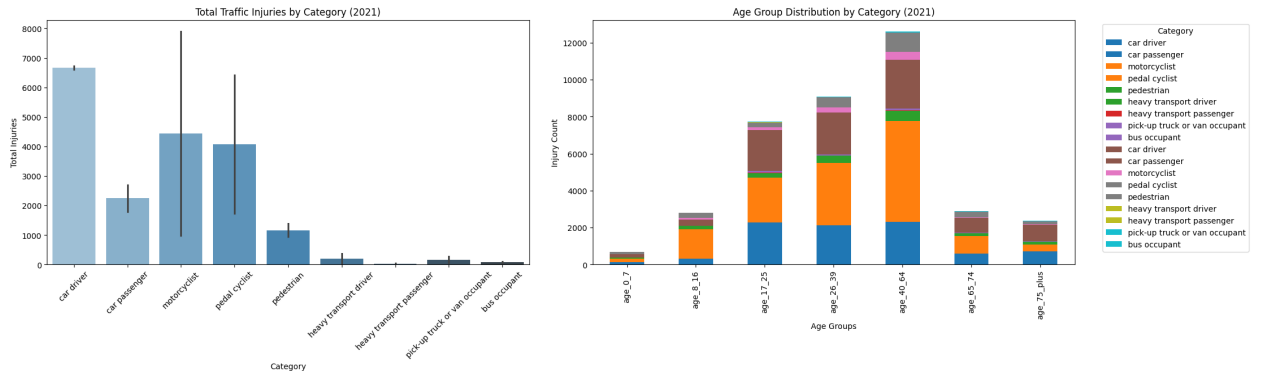


Figure 15: Traffic Injuries by Category and Age Group in 2021

Proportion of Total Injuries by Category (2021)

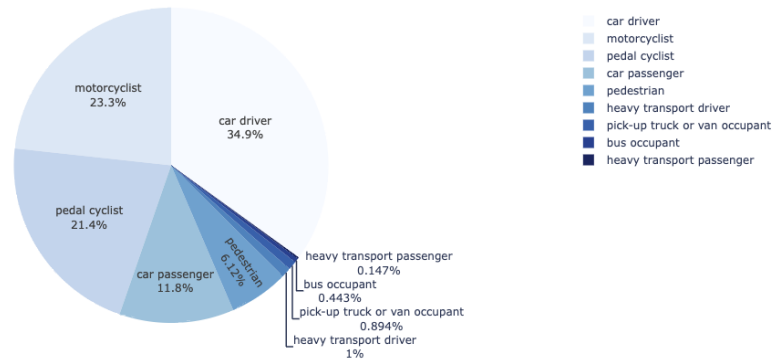


Figure 16: Proportional Distribution of Traffic Injuries by Category (2021)

4.2.2 Traffic busy level versus music listeners

We have observed that many drivers enjoy listening to music while driving. This led us to investigate whether busy traffic conditions impact the popularity of music tracks. To explore this, we set up a music data harvester alongside our real-time traffic harvester in VIC.

The line chart in Figure 17 compares daily music listener increments with average traffic delay (scaled by 1000) over a 7-day period. A Pearson correlation of 0.30 indicates a weak positive relationship, with occasional alignment in trend.

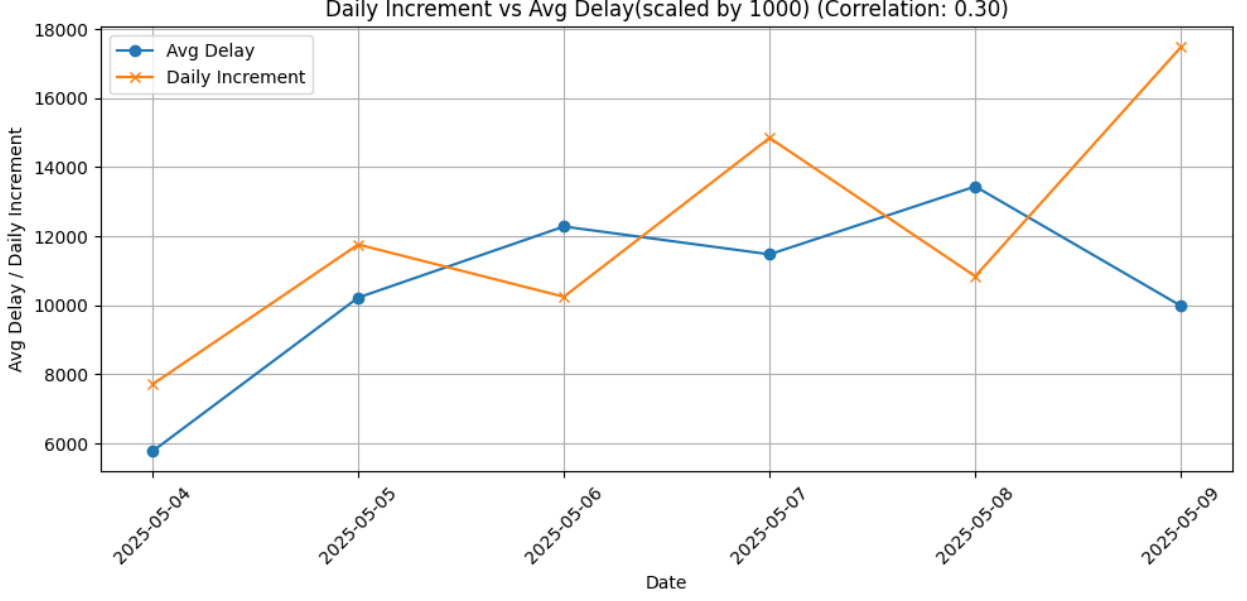


Figure 17: Daily music listener increments and average traffic delay (scaled by 1000) from May 4 to May 9, 2025.

4.2.3 Connection with YouTube video data

To investigate traffic-related patterns in urban life, YouTube video metadata was collected using lifestyle-related keywords specific to Melbourne. The search focused on terms such as “*Melbourne vlog*”, “*Melbourne cafe*”, and “*Melbourne food*”, aiming to capture content related to personal travel, dining experiences, and local exploration. The dataset includes video titles, descriptions, timestamps, locations (if available), and popularity metrics such as view counts. This collection provides insights into how social media activity reflects patterns of movement and congestion in urban areas.

Figure 18 provides an overview of the Melbourne lifestyle video data over time. The top panel shows daily upload and total view counts, highlighting periodic fluctuations and a general upward trend in content production from mid-March to early May. To quantify the relevance and popularity of each lifestyle-related YouTube video, we define two complementary scoring metrics: **Score1** and **Score2**.

Score1 is a *time-sensitive popularity score*, designed to reflect both the absolute engagement of a video and its temporal freshness. It is calculated as a weighted sum of view count, like count, comment count, and favourite count, followed by an exponential decay based on the number of hours since publication. This ensures that recently popular videos receive a higher score, making **Score1** suitable for identifying trending content in near real-time.

Score2, in contrast, is a *normalised engagement score* that focuses on the quality of audience interaction rather than the raw volume. It is computed as a weighted ratio of likes, comments, and favourites relative to views, placing greater emphasis on user responses per view rather than total exposure. This design makes **Score2** more stable over time and less biased toward viral content with high views but low interaction. The raw ratio is cut according to the quantiles.

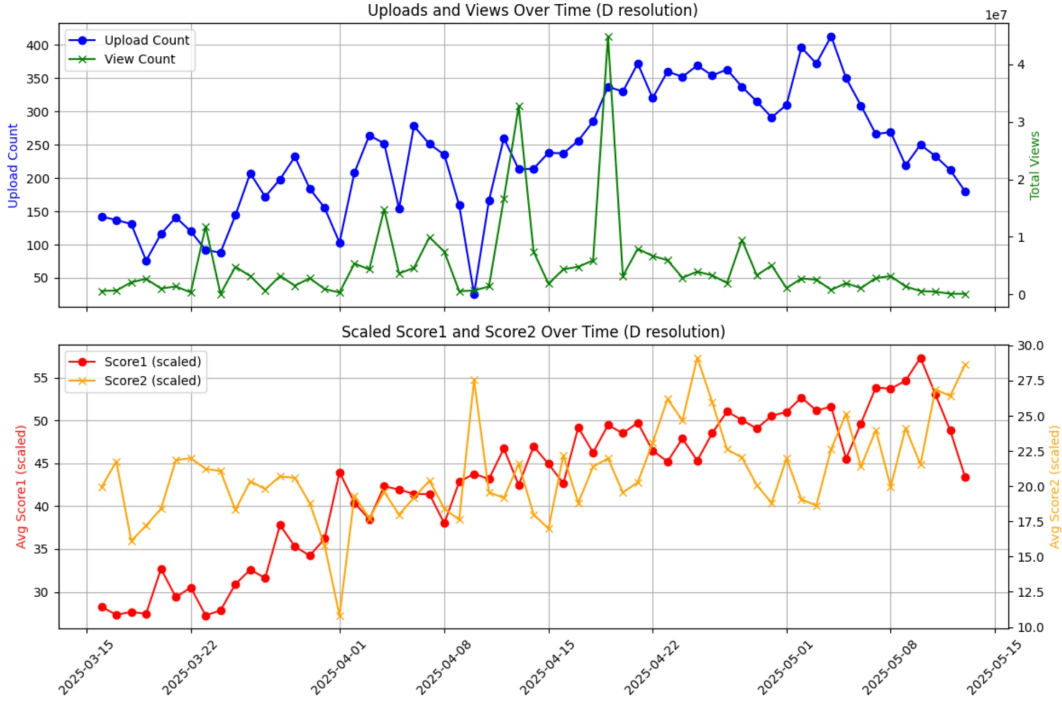


Figure 18: Youtube Data Overview

Both scores are scaled to the range $[1, 100]$ for better interpretability. By calculating these two scores, we are able to capture different aspects of content dynamics: **Score1** highlights videos that are currently gaining attention at scale, and **Score2** identifies videos with a high degree of viewer engagement, regardless of reach.

The bottom panel in Figure 18 displays the average values of two sentiment or engagement-related metrics (**Score1** and **Score2**). While **Score1** shows a relatively steady increase, **Score2** exhibits higher volatility. Together, these trends offer insights into how lifestyle-related video activity and public engagement evolved during the observation period.

To explore the relationship between online sentiment and traffic conditions, we aligned YouTube video scores with traffic congestion data using a common timestamp. Video scores were averaged in 2-hour intervals and merged with traffic data for comparison.

We computed Pearson and Spearman correlations between scaled sentiment scores and average congestion. The results showed weak correlations with no statistically significant linear or monotonic relationships. This suggests the connection may be weak, non-linear, or not well-captured by the current scoring approach. Further analysis using lagged variables or alternative metrics may provide deeper insight.

In order to capture potential delayed effects between online sentiment and real-world traffic flow, we applied a lagged Pearson correlation analysis to the scaled video scores. For both **score1_scaled** and **score2_scaled**, values were shifted by time lags ranging from -2 to $+2$ intervals (i.e., -4 to $+4$ hours, given the 2-hour aggregation window), and correlated with the corresponding congestion levels.

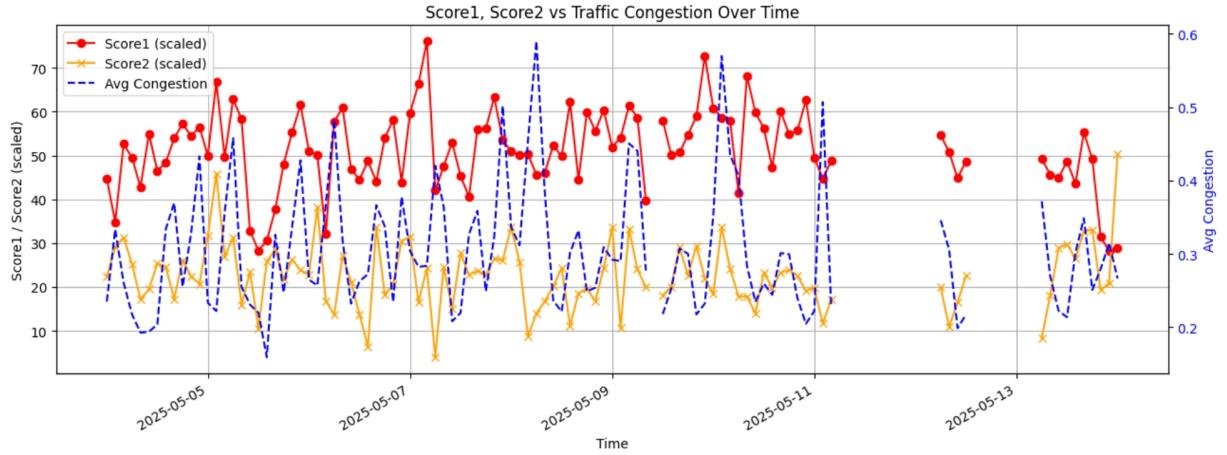


Figure 19: Temporal comparison of YouTube sentiment scores (**Score1**, **Score2**) with average traffic congestion.



Figure 20: Lagged comparison of scaled YouTube sentiment scores (**Score1** and **Score2**) with average traffic congestion.

As shown in Figure 20, **Score1** reveals a weak but noticeable trend that spikes in video popularity are often followed by increased traffic congestion after a short delay. This pattern suggests that bursts of online attention, measured by overall views, likes, and comments, may precede short-term rises in congestion.

In contrast, **Score2**, representing normalised engagement quality, shows no consistent relationship with traffic conditions, indicating limited relevance to real-world patterns.

These findings imply that raw popularity trends (**Score1**) may have some limited predictive value for short-term congestion patterns, whereas normalised interaction quality (**Score2**) does not. Nonetheless, the observed correlations are still moderate at best, and further exploration, such as lagged Granger tests or non-linear modelling, may be necessary to fully understand the underlying dynamics.

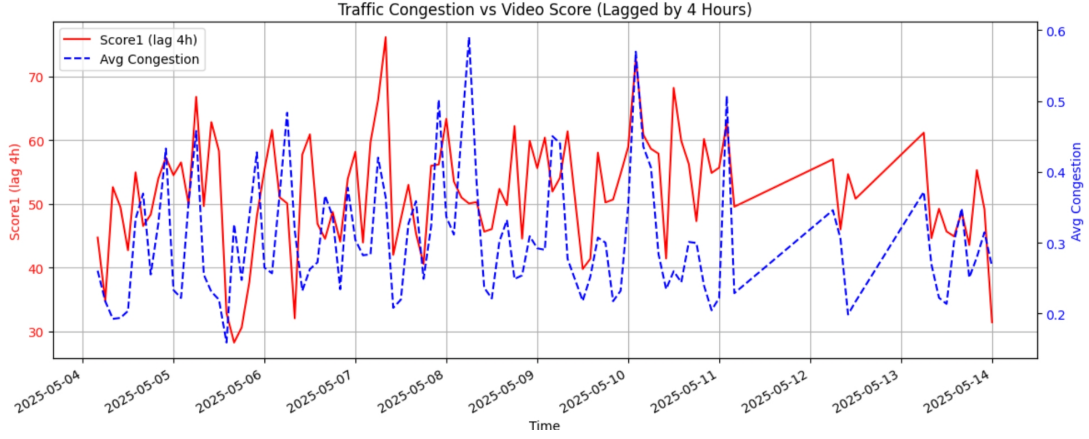


Figure 21: Comparison of traffic congestion and `Score1` (lagged by 4 hours).

Figure 21 shows the time series of `score1_scaled` shifted forward by 4 hours alongside average traffic congestion levels. A visual comparison reveals that peaks in video popularity often precede corresponding rises in congestion, reinforcing the modest positive correlation observed at this lag in the statistical analysis.

Taken together, these results suggest that spikes in video popularity, particularly those that occur shortly before rises in congestion, may serve as weak yet timely indicators of traffic pressure. While not definitive, such patterns highlight the potential of leveraging online attention dynamics as an early-warning signal for real-world mobility trends.

4.3 Hot-word Analysis Before and After the Australian Federal Election

4.3.1 Hot-word Analysis of Australian Federal Election Data on Reddit

We used Reddit’s API to scrape posts from subreddits related to the Australian federal election, along with all comments under those posts. From this dataset, we conducted a hot-word analysis to identify the most frequently discussed terms during different election cycles. Figure 22 shows hot-word frequencies across four selected time periods.

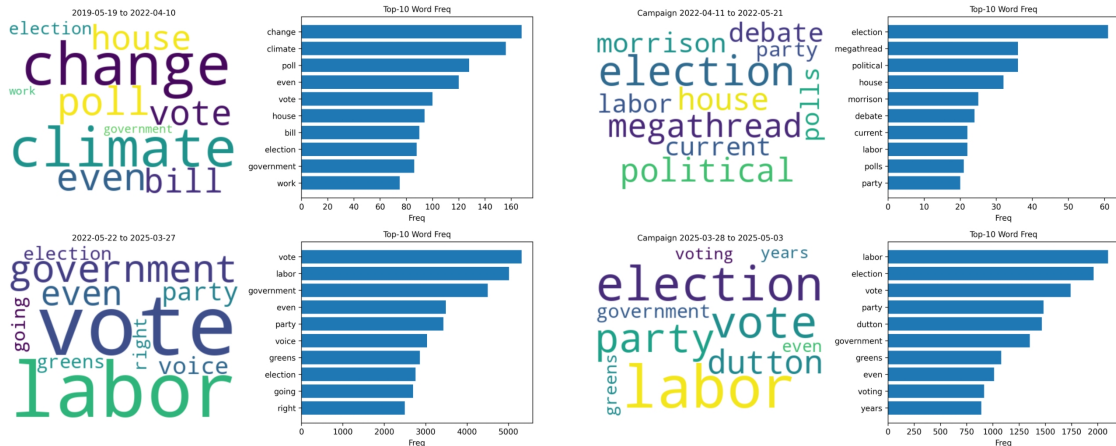


Figure 22: Hot Word Trends Across Four Key Periods (2010–2025)

Between May 2019 and April 2022, climate and policy-related terms, such as *change*, *climate*, and *bill*, appeared frequently, while explicit political terms were less frequent. Since 2022, baseline political discussion has remained elevated, blurring the line between campaign and non-campaign periods. The April 2022 campaign saw a surge in terms like *election*, *labour*, *polls*, and Reddit-specific words like *megathread*. This heightened engagement persisted into 2025, with *vote*, *party*, *government*, *voice*, and *greens* remaining prominent.

During the 2025 campaign, recurring terms like *election* and *vote* remained dominant, while new entries like *dutton* also rose in frequency.

Based on posts and comments related to federal elections that were scraped from Reddit between 2010 and 2025, we divided the timeline into campaign and non-campaign periods and plotted a line chart of the top 10 hotwords and their frequencies.

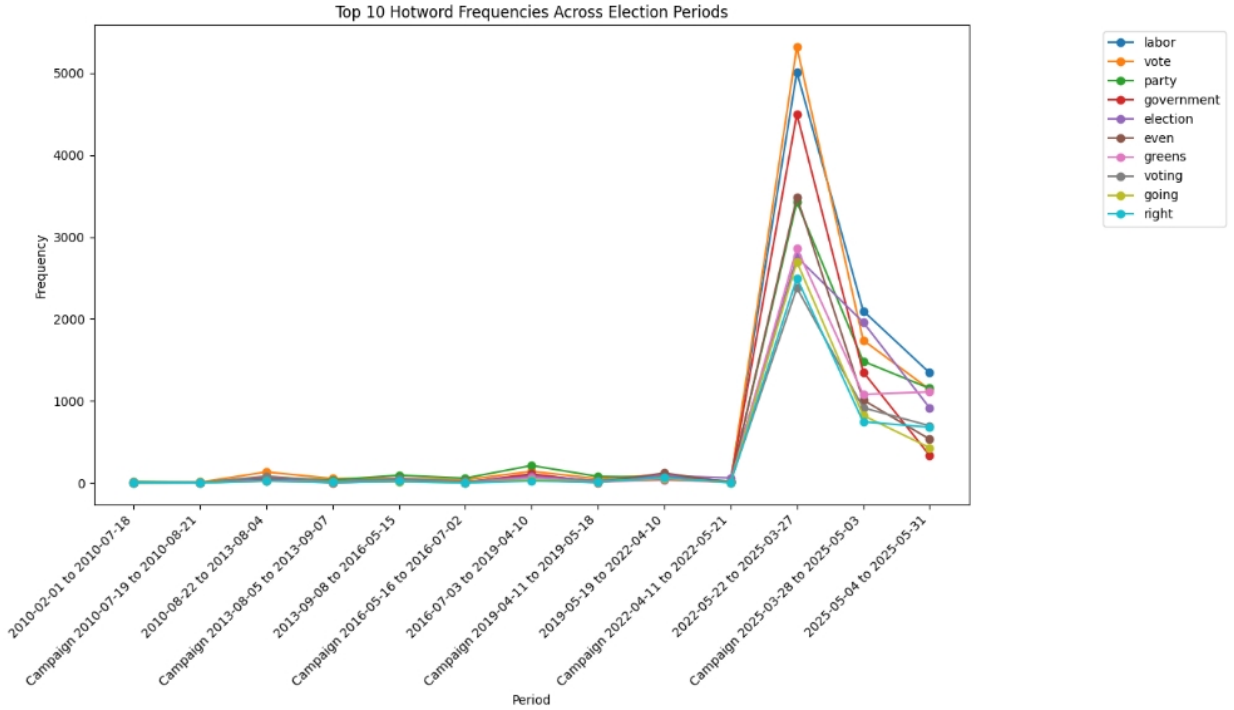


Figure 23: Top 10 Hot Word Frequencies Over 15 Years of Election Periods

From Figure 23, it is evident that election-related discussions on Reddit were minimal in earlier years, with significantly lower hot-word frequencies. Political terms consistently explode in frequency during campaign periods.

Reddit’s discourse follows a clear two-phase pattern: policy topics dominate between elections, while campaign periods shift attention to voting, party names, and political figures. Since 2022, the baseline level of election talk has stayed high, blurring the line between campaign and non-campaign periods.

4.3.2 Hot-word Analysis of Australian Federal Election Data on YouTube

In this section, we conduct a hot-word analysis of YouTube video data related to Australian federal elections. By extracting and tracking high-frequency keywords from video titles, descriptions,

and tags, we aim to uncover dominant topics and public discourse trends across different election periods. The analysis leverages preprocessed text data, with stopwords and non-informative tokens removed, to focus on meaningful content. We group the data by campaign and non-campaign windows, and visualise keyword distributions using word clouds. This approach provides insights into how public attention shifts around election events and highlights the most salient themes in political video content over time.

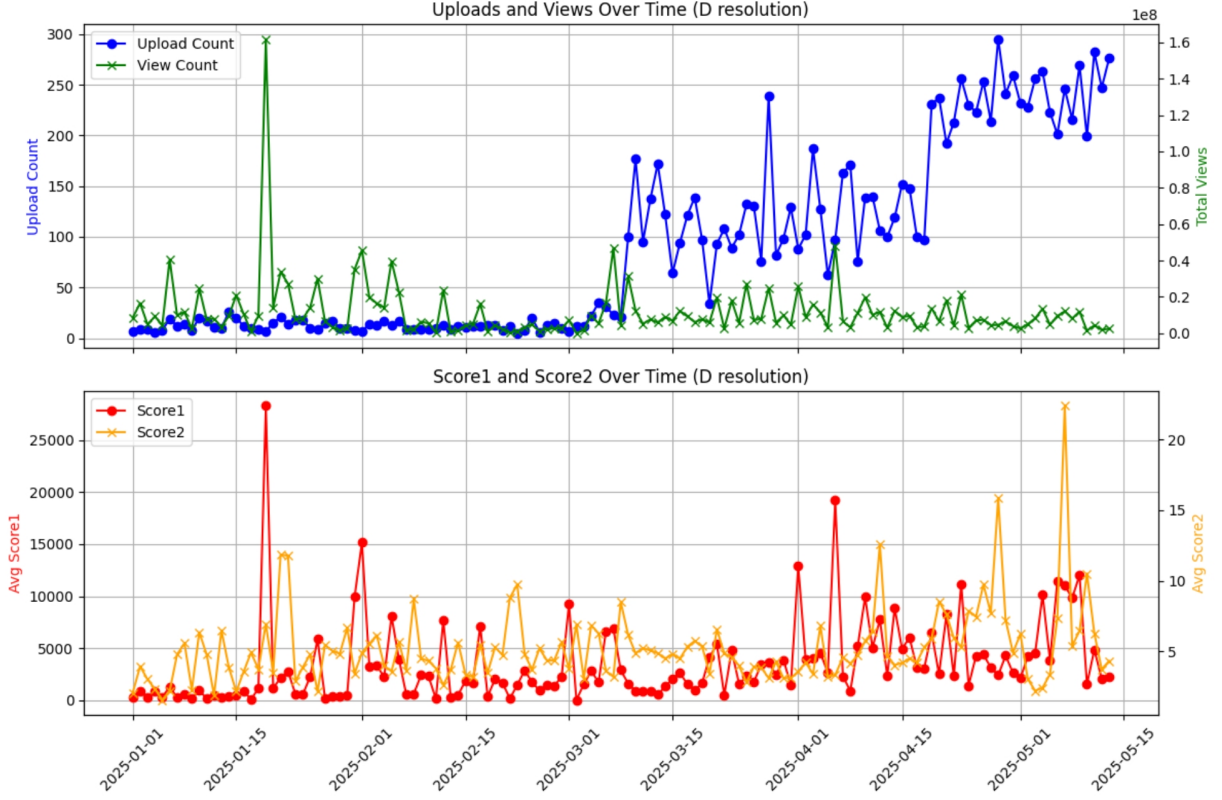


Figure 24: Daily YouTube election video data from January to May 2025.

Figure 24 summarises YouTube activity and sentiment trends related to the Australian federal election. The top subplot shows the daily upload count (blue) and total view count (green) of videos. The top subplot shows daily uploads (blue) and views (green), with a clear rise in uploads starting March 2025 and earlier view spikes indicating viral or event-driven content. The bottom subplot presents **Score1** (red) and **Score2** (orange), capturing popularity-weighted and engagement-normalized sentiment. **Score1** shows sharp, short-term spikes, while **Score2** remains steadier with a slight increase during the campaign, reflecting changing viewer interaction and content reach.

To identify prominent themes in election-related YouTube content, we performed a weighted hot-word extraction based on video metadata. Titles, descriptions, and tags were combined into a single text field for each video, then cleaned by removing punctuation, links, and common stop words. Each video was assigned a weight based on its scaled engagement score, reflecting normalised viewer interaction. The cleaned text was duplicated proportionally to this weight, allowing higher-impact videos to contribute more heavily to overall word frequencies. The top keywords were then visualised using a word cloud to highlight their relative prominence.

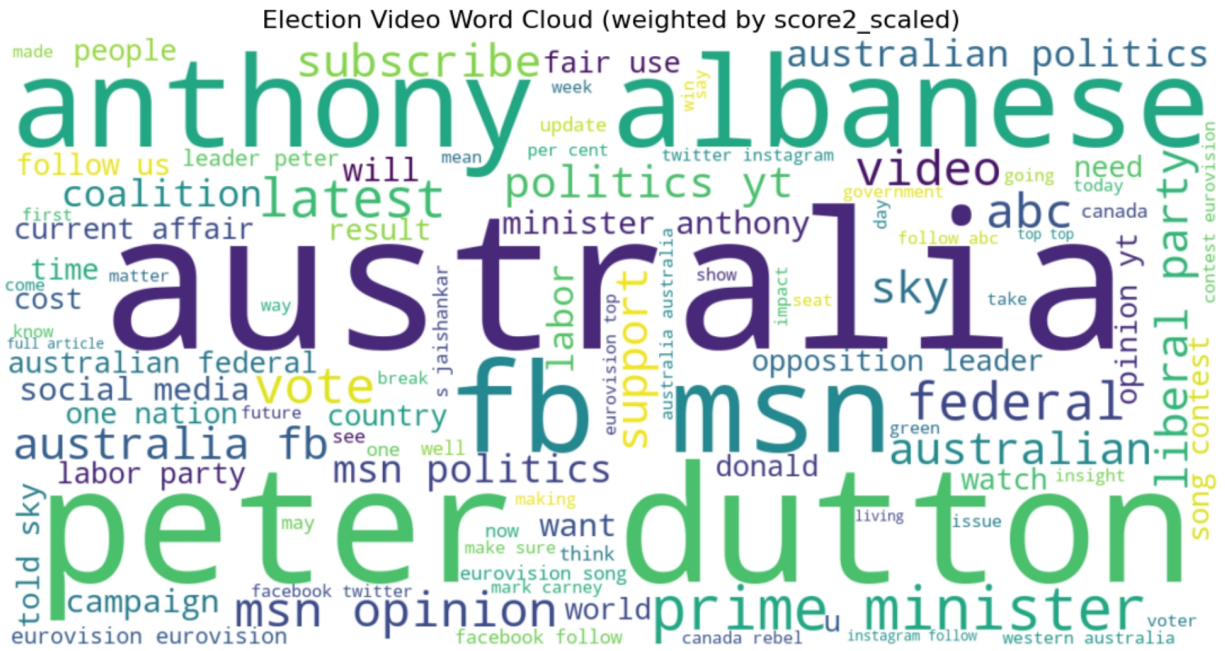


Figure 25: Word cloud of frequently occurring terms in Australian election-related YouTube videos, weighted by normalised engagement (`score2_scaled`).

As shown in Figure 25, the most prominent terms include political figures like *peter dutton* and *anthony albanese*, alongside platform-related terms such as *fb*, *msn*, and *sky*. Keywords like *vote*, *prime minister*, and *coalition* highlight strong alignment with electoral themes. These hot words reflect the dominance of political and media narratives during the campaign period.

4.3.3 Temporal Hot-word Comparison Across Key Campaign Phases

To examine how keyword salience changed throughout the 2025 Australian federal election cycle, we divided the YouTube video dataset into four key periods: **before March 2025** (pre-campaign baseline), **March 2025** (early campaign activity), **April 2025** (intensive campaign phase), and **after May 2025** (post-election discussion).

This segmentation aligns with the official campaign timeline, where March marks the pre-write escalation and May 3rd serves as the polling date. For each period, we generated separate word clouds weighted by `score2_scaled` to reflect engagement levels, allowing us to capture shifting public focus and evolving discourse patterns in election-related content.

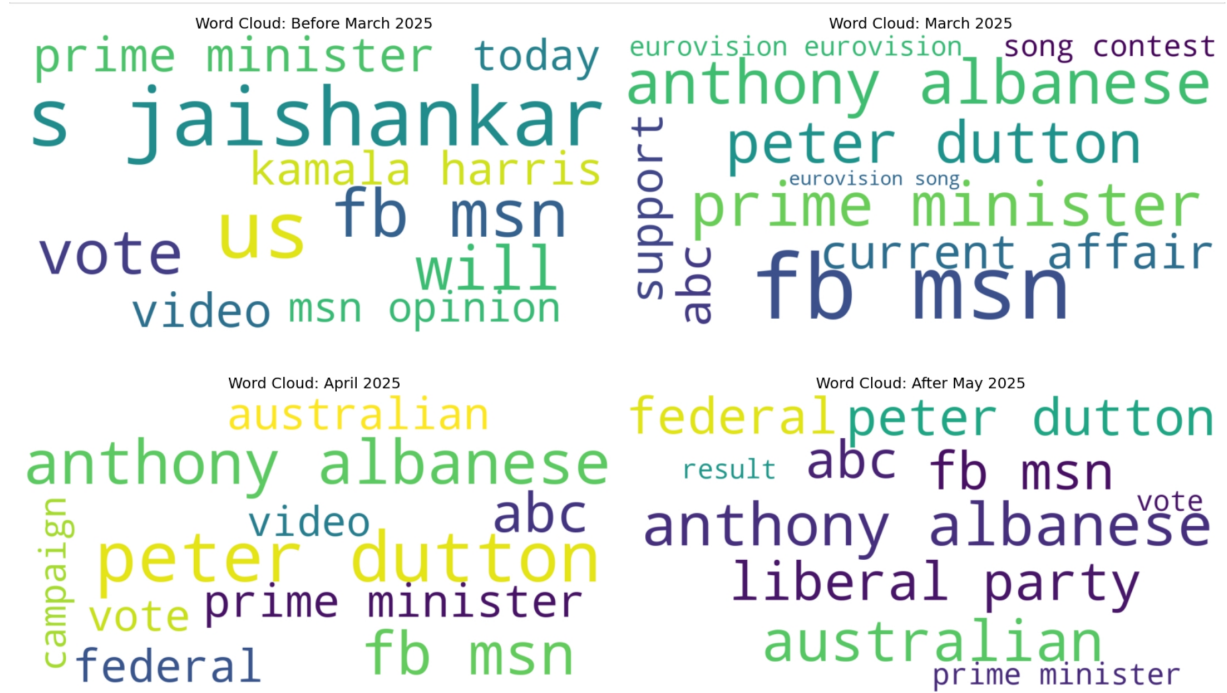


Figure 26: Word clouds showing the most frequent terms in YouTube election-related videos across four phases.

As shown in Figure 26, clear thematic transitions emerge across the four phases. Before March, foreign political figures such as *S. Jaishankar* and *Kamala Harris* unexpectedly dominate. Although our data collection focused on Australian federal election content, YouTube’s recommendation and tagging systems may have introduced loosely related global political videos. This suggests that early discussions carried a broader geopolitical lens, potentially amplified by algorithmic cross-linking and shared political tags.

In March, domestic figures like *Anthony Albanese* and *Peter Dutton* become more prominent, along with terms such as *prime minister* and *eurovision*, indicating a mix of political and cultural discourse. April’s word cloud reflects campaign maturity, with strong emphasis on terms like *vote*, *federal*, and leadership names, consistent with intensified electoral messaging.

After May, the discourse shifts toward post-election analysis, with terms such as *result*, *liberal party*, and *australian* indicating increased focus on political outcomes and party-specific reflection.

4.4 The Impact of the Australian Federal Election

4.4.1 The Australian Federal Election’s Impact on Inflation

After the September 2013 election, the carbon tax was repealed and took effect in July 2014. This led to a noticeable drop in energy prices, including electricity, which in turn lowered the inflation rate in mid-2014.

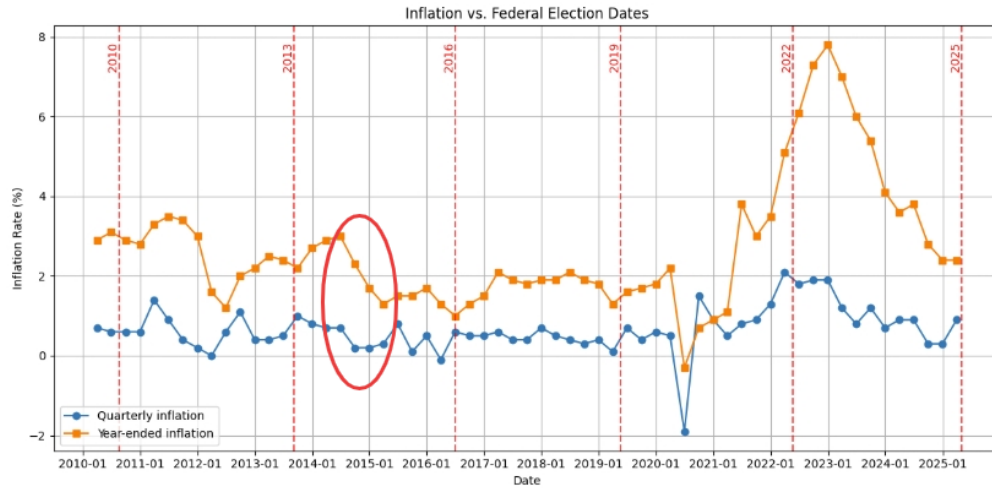


Figure 27: Inflation rates over time in relation to Australian federal election dates.

From the above, it can be seen that election day does not directly impact inflation. The actual inflation trend is driven by larger factors such as the COVID-19 pandemic, global commodity prices, supply-chain dynamics, and central bank interest-rate decisions. For instance, structural reforms like large-scale tax reforms (the carbon tax repeal) can produce short-term inflation fluctuations, but they typically roll out over several months or even years after the election, rather than being triggered on election day itself.

4.4.2 The Australian Federal Election's Impact on House Prices

In Figure 28, we plotted Australia's house-price trends from 2010 to 2025 and segmented them by federal election periods to examine potential links between elections and housing dynamics.

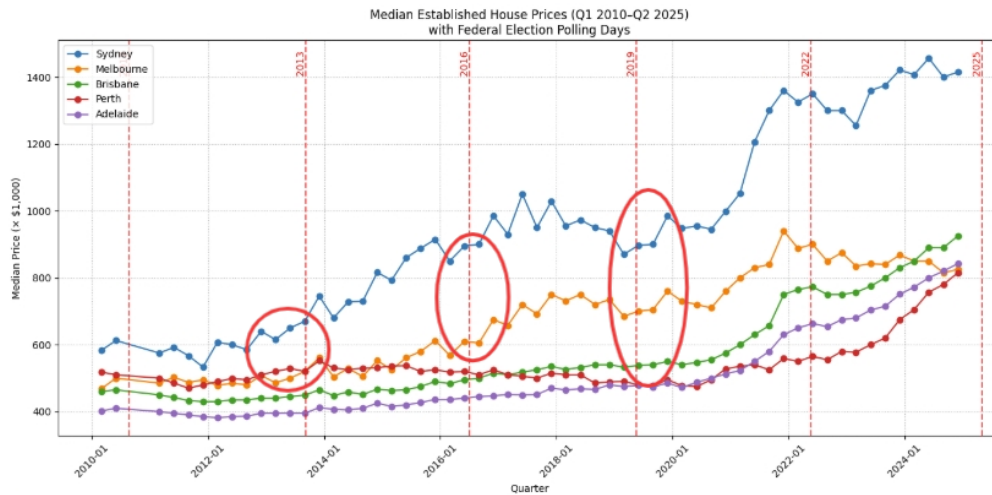


Figure 28: Median house prices in major Australian cities from 2010 to 2025, overlaid with federal election dates.

Prior to the 2013 election, price growth slowed amid concerns that Labor's carbon tax would raise energy and manufacturing costs. After the Coalition won and repealed the tax in mid-2014,

confidence returned and prices rebounded, particularly in Sydney and Melbourne. A similar pattern occurred around the 2016 and 2019 elections, where speculation over potential changes to negative gearing and capital gains tax dampened growth. In both cases, price momentum recovered once those reforms failed to realise.

In contrast, the 2021–22 housing boom was largely driven by pandemic-era stimulus measures and record-low interest rates, rather than election outcomes. When the Reserve Bank began raising rates after mid-2022, demand cooled, and prices entered a correction.

Overall, while elections may trigger short-term “slowdown–rebound” swings in investor sentiment, long-term housing trends are more strongly shaped by interest rates, fiscal stimulus, and supply–demand fundamentals than by electoral outcomes.

4.5 Inflation

4.5.1 Synchrony between Reddit Salience and Official CPI

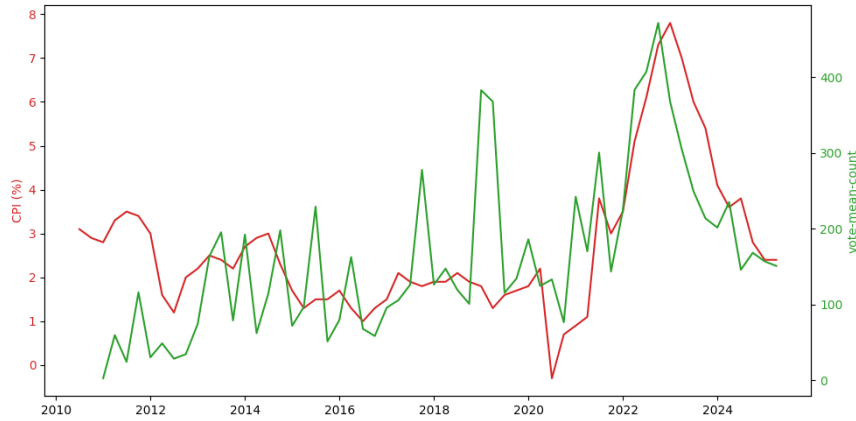


Figure 29: Quarterly CPI (left axis, red) versus mean Reddit vote for inflation-related posts (right axis, green), 2010–2025.

To assess whether online discourse can reflect macroeconomic conditions in real time, we compared the year-ended Consumer Price Index (CPI, %) published quarterly by the RBA with a social media proxy for inflation salience. This proxy is defined as the quarterly average vote score of Reddit submissions whose titles or self-texts contain inflation-related keywords. After removing posts by deleted authors and obvious duplicates, individual post scores were aggregated to compute the quarterly mean. Both time series were resampled to calendar quarter-end over the period 2010 Q1–2025 Q1: CPI values are reported on a quarterly basis, while the Reddit metric preserves the raw communal endorsement signal.

Figure 29 displays the two series on dual y -axes. Visually, the steep rise in CPI during the 2021–2022 inflation episode is mirrored by a pronounced surge in Reddit vote averages, followed by a synchronised decline in late 2023. A Pearson correlation coefficient of $r = 0.55$ over the full sample indicates a moderate yet economically meaningful positive association. The turning points of both series occur within the same quarter, suggesting that collective attention on the platform responds to official price prints with minimal lag.

In sum, average Reddit post votes serve as a timely barometer of public concern about inflation.

This finding justifies the inclusion of Reddit engagement metrics as an early-warning layer in our analytical pipeline.

4.5.2 Core Topics of Reddit “Inflation” Posts

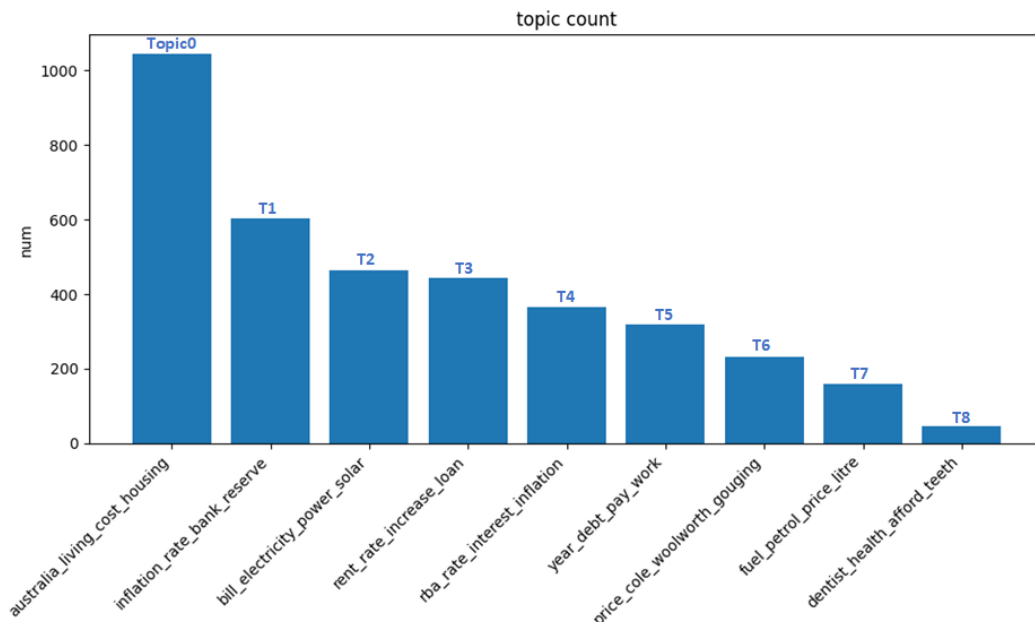


Figure 30: Top Reddit topics related to inflation, based on keyword groupings and frequency counts.

To uncover the principal subtopics that capture public attention under the “inflation” label, we conducted topic modelling on the combined titles and self-texts of Reddit submissions collected between 2010 and 2025. Preprocessing followed the pipeline outlined in Section 3.3, after which we fitted a BERTopic model in English mode specifying ten topics. The automatically detected noise topic (label -1) was discarded, yielding nine interpretable clusters.

Figure 30 displays the relative frequency of each topic label, while Figure 31 presents the top-weighted representative terms for each cluster.

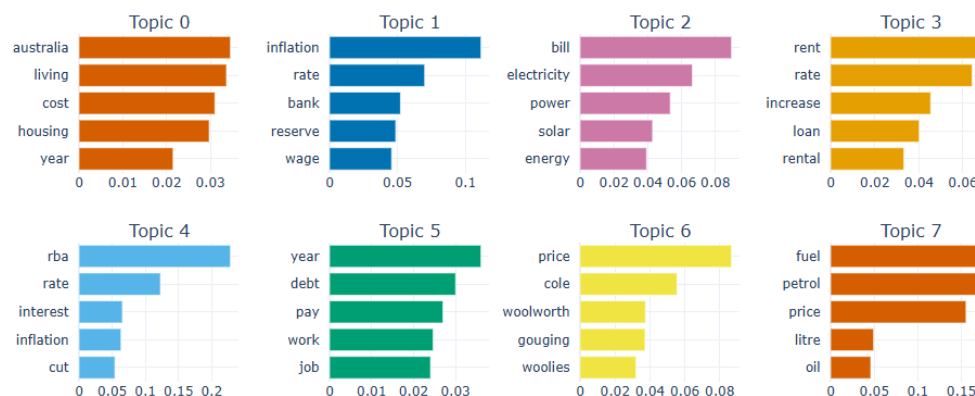


Figure 31: Representative keywords for the main topics

The similarity matrix (Figure 32) demonstrates the relationship between topics. The strongest associations are among the cost-of-living, rent, and debt-related topics. The dimensionality-reduced scatter plot (Figure 33) further validates this spatial arrangement.

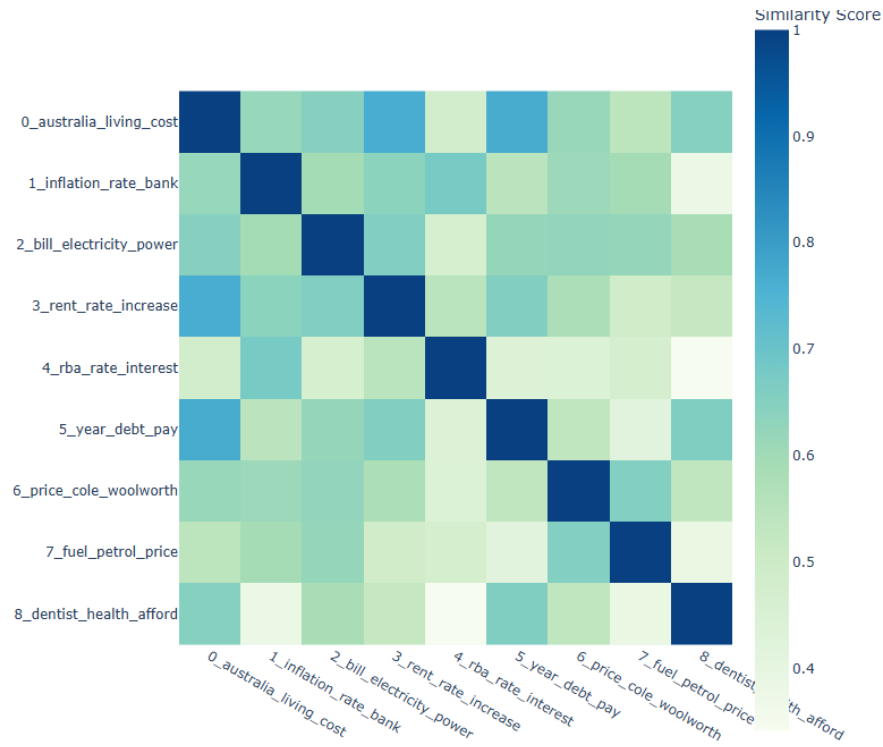


Figure 32: Semantic similarity matrix between topics

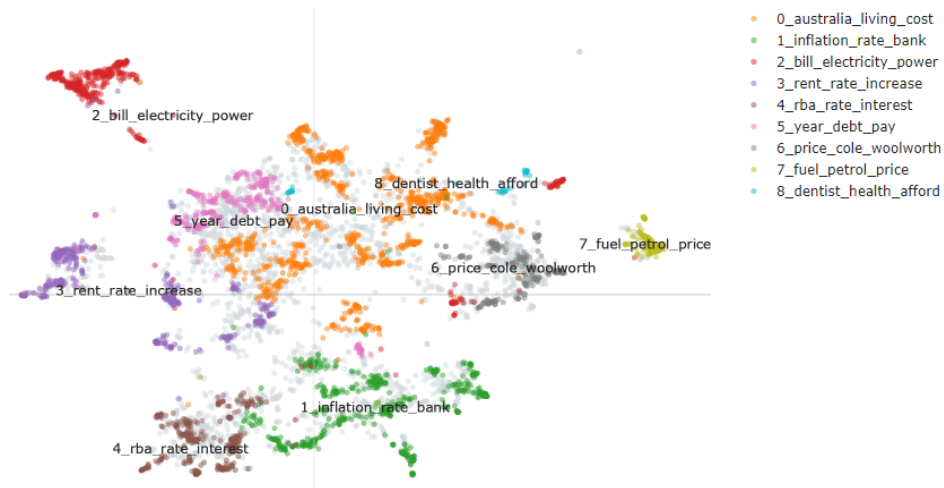


Figure 33: Topic distribution of posts in the two-dimensional embedding space

The time-series visualisation (Figure 34) shows how inflation-related topics evolved over time. From the second half of 2022 onward, the curves for *cost_of_living*, *ren_property*, and *tax_debt_inflation* rise almost in parallel. This supports the intuitive observation that Australia's inflationary shock extended into housing, taxation, and personal finance. At the same time, the *RBA_interest_rate*

and *supermarket_pricing* topics display oscillating trends that align with official rate hikes and fluctuations in media coverage.

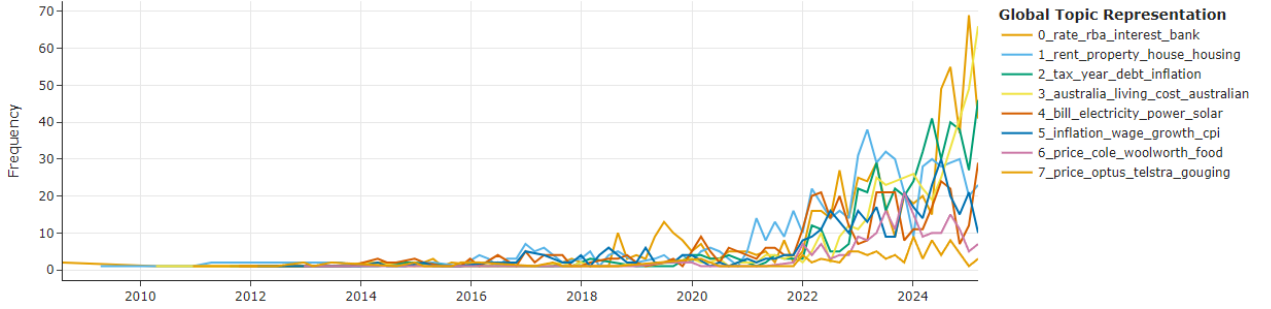


Figure 34: Relative frequency of topics over time (2010–2025)

However, due to functional limitations of the data-retrieval API, post coverage in the early years is sparse, and consequently no reliable conclusions about topic prevalence prior to 2018 can be drawn. Therefore, further investigation is required to reveal the broader context.

4.5.3 Inflation and crime

We combine the RBA’s year-on-year CPI growth with ABS victim counts for 12 offence categories (1993–2023). Each series is first resampled to calendar-year means and then transformed to percentage change, ensuring that all variables represent comparable growth rates rather than raw levels.

Figure 35 plots the Pearson correlation between CPI growth and the growth rate of each offence. Contrary to the naïve “high inflation → more crime” intuition, only a subset of crimes displays a clear positive relationship with CPI. *Blackmail/Extortion* tops the list ($\rho \approx +0.54$), followed by moderate links for *Murder*, *Sexual Assault*, *Unlawful Entry with Intent*, and *Motor-Vehicle Theft* ($\rho \approx +0.25$ – 0.33). In contrast, robbery-type offences (*Armed*, *Unarmed*, and generic *Robbery*) are negatively correlated ($\rho \approx -0.35$), while *Attempted Murder* and *Kidnapping/Abduction* hover around zero.

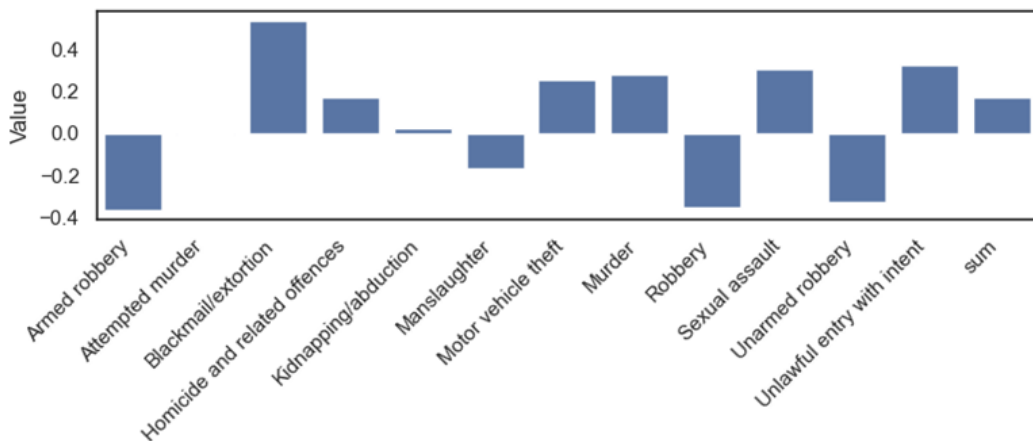


Figure 35: Correlation between CPI growth and growth in victim counts (1993–2023). Positive bars indicate crimes that rise with inflation; negative bars indicate the opposite.

Among all crime categories, *Blackmail/Extortion* shows the highest correlation with CPI ($\rho \approx +0.54$). The two 0–1 normalised line plots in Figure 36 oscillate almost in unison. Periods of high inflation coincide with pronounced increases in the number of blackmail and extortion victims, supporting the hypothesis that economic pressure may exacerbate incentives for coercive, profit-driven crimes.

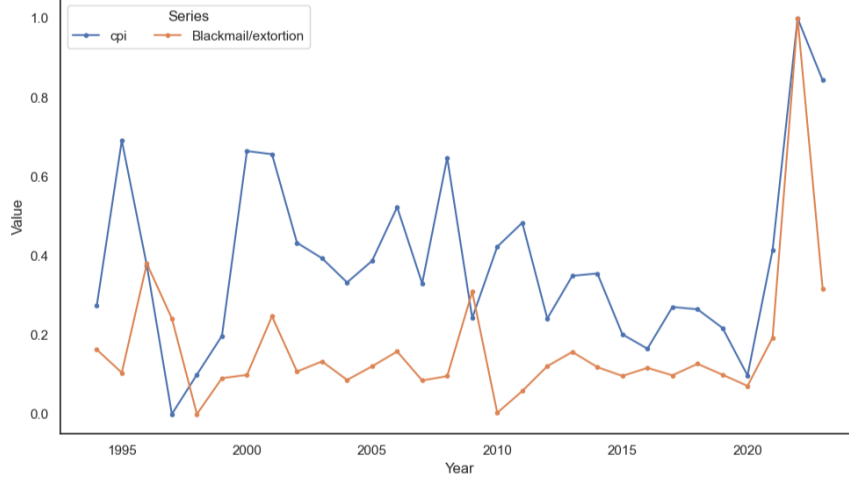


Figure 36: Min-max-normalised CPI growth vs. Blackmail/Extortion victim growth. The two series rise and fall in tandem, yielding the highest positive correlation observed.

However, “more inflation, more crime” fails in the aggregate. When all 12 categories are summed (Figure 37), the overall victim count shows only a weak positive correlation with CPI ($\rho \approx +0.17$). Positive and negative components offset each other, masking heterogeneous responses at the category level. This underscores that inflation is *not* a universal driver of crime; its influence is conditional on the economic incentives and opportunity structures specific to each offence type.

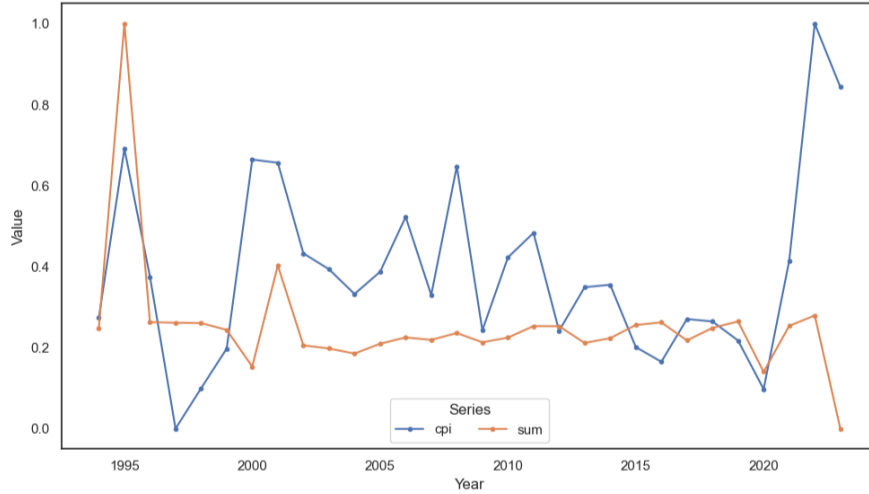


Figure 37: CPI growth vs. aggregate victim growth (*sum*). The muted association demonstrates that macro-level crime trends cannot be inferred from price dynamics alone.

Therefore, the criminological impact of inflation is heterogeneous: crimes associated with *economic stress or leverage* (e.g., extortion, break-ins, and certain violent acts) exhibit a positive co-movement

with CPI, whereas offences driven by *street-level opportunity and policing intensity* (e.g., robberies) tend to move inversely or show no systematic relationship. Consequently, policy analysis should incorporate offence-specific mechanisms—such as risk–reward calculus and law-enforcement pressure—rather than assume a blanket “inflation breeds crime” narrative.

5 Evaluation and Discussion

5.1 System performance and reliability

Our system is designed for high performance using a multi-node Kubernetes cluster architecture, consisting of one master node and three worker nodes. The three worker nodes enhance reliability, ensuring that even if one node fails, the master node can continue to distribute workloads across the remaining two nodes. However, a potential weakness lies in the master node itself, which serves as a single point of failure. If the master node becomes unavailable, the Kubernetes control plane will be disrupted, preventing API management, workload scaling, and the scheduling of new pods. This could result in a complete system outage, as no new workloads can be deployed and existing ones cannot be effectively managed. We believe that adopting a multi-master node architecture would help mitigate this risk by ensuring high availability of the control plane and maintaining cluster operations even if one master node fails.

Our Elasticsearch uses a dual-master configuration, where both nodes also serve as data nodes with 100 GiB of storage each. This setup ensures efficient data indexing and querying, enabling fast read and write operations. In the event of a failure in one Elasticsearch master node, the other node can continue to manage data and maintain service availability without interruption.

Fission is used to deploy scalable serverless functions, with multiple harvesters and ReST APIs distributed across the worker nodes. This setup supports parallel data processing and efficient request handling while offering fault tolerance. If a node fails, functions can automatically restart on another available node, minimising downtime and ensuring continuous operation.

Furthermore, a Redis caching layer between the front end and ReST APIs significantly enhances performance by reducing response times for frequent API calls, offloading traffic from the backend, and minimising redundant queries to Elasticsearch. It also adds an extra layer of reliability by allowing the system to continue serving cached data even if Elasticsearch becomes temporarily unavailable.

5.2 Scalability Analysis

Our project is also designed with strong scalability, leveraging Fission’s serverless capabilities. When a new data source needs to be added, a new harvester function can be easily created and deployed without altering the existing infrastructure. Since all harvesters operate independently, the system can scale horizontally by simply adding new functions as needed. These harvesters are automatically scheduled across available worker nodes based on resource availability. Combined with the elasticity of Fission and the distributed nature of our architecture, the system can dynamically adapt to increased workloads or new data streams with minimal configuration overhead.

5.3 Limitations

Despite the strengths stated above, our system has some limitations.

5.3.1 Sentiment Analysis Accuracy

The algorithm we employ lacks sufficient precision. It cannot reliably detect sarcasm, irony, or nuanced idioms common on social media, which may lead to subpar classification and data-analysis results.

5.3.2 Data Sparsity

Some Australia-related subreddits see low posting volumes outside election peaks. Small sample sizes introduce high variance in word-clouds and sentiment trends.

Figure 38 illustrates the monthly post volume in the `mastodon-144g` dataset on a logarithmic scale. Two notable surges are evident. The first increase, beginning in November 2022, aligns with the acquisition of Twitter by Elon Musk [16]. This triggered a widely reported migration of users—particularly journalists, academics, and tech communities—to alternative platforms like Mastodon. The second, and much more dramatic, spike in post volume occurs at the end of 2024. This surge likely corresponds to the global online discourse surrounding the 2024 United States presidential election, a high-profile event that drew significant attention on decentralized platforms such as Mastodon. However, the sudden drop in activity observed in early 2025 is not indicative of a real decline in platform usage. Instead, it reflects a lack of complete data harvesting for recent months, resulting in artificially low post counts. These shifts highlight how both external sociopolitical events and internal pipeline factors can significantly influence temporal data availability and interpretation in social media analytics. This discrepancy in data volume poses challenges for temporal analyses, as periods with higher post density can disproportionately influence results such as word frequency and sentiment. Consequently, interpretations of longitudinal trends must consider this structural imbalance in the dataset.

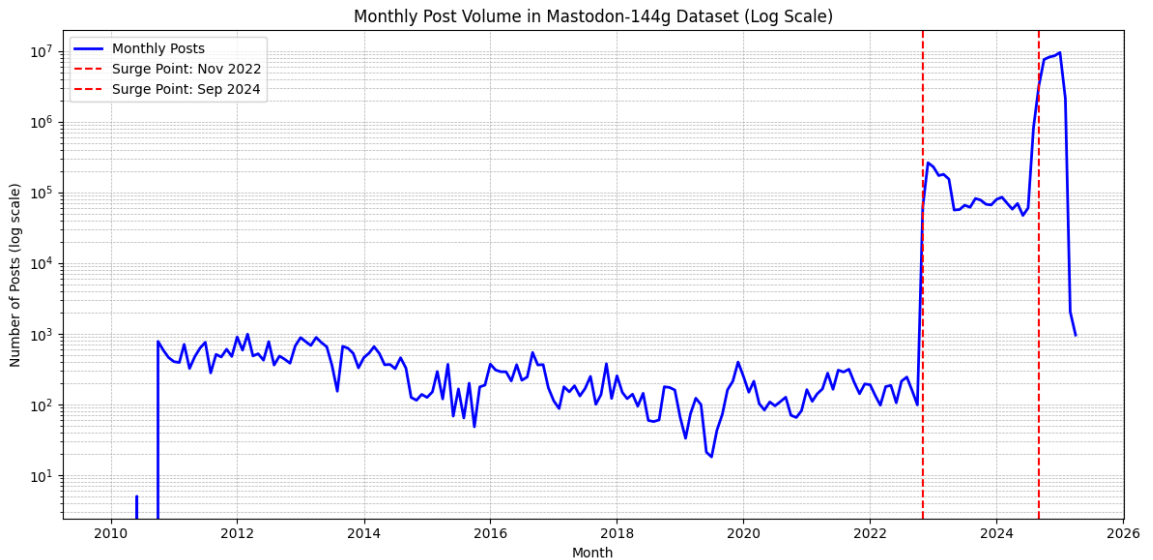


Figure 38: Monthly post volume in the `mastodon-144g` dataset from 2010 to 2025 (logarithmic scale)

5.3.3 Inaccurate API Search and Recommendation

One notable limitation of our data collection lies in the inherent ambiguity of the YouTube API’s search and recommendation mechanisms. Although we explicitly queried for Australian federal election-related content, the API results often included videos that were only loosely related or entirely unrelated to the intended topic. For instance, high-view advertisement-style videos, which are algorithmically favoured due to their popularity or promotional status, frequently appeared among the top results. The search occasionally returned content about elections in other countries, such as the United States or India, likely due to shared keywords like “election” or trending political figures. This introduces noise into the dataset and may dilute the focus on Australian political discourse.

While efforts were made to mitigate such inaccuracies through keyword tuning and post-filtering, the nature of YouTube’s algorithm-driven visibility and metadata matching remains a source of uncertainty in the representativeness of the data.

5.3.4 API Rate Limits and Latency:

Harvesting large volumes of Reddit or Mastodon posts can hit rate limits, causing ingestion delays.

5.4 Discussion on pros and cons of Kubernetes, Fission, ElasticSearch

Technology	Pros	Cons
Kubernetes	• Self-healing pods and auto-rescheduling • Declarative YAML/JSON, GitOps-friendly	• Steep learning curve (many new concepts) • Complex CNI networking & RBAC setup
Fission	• Per-function autoscaling • Built-in HTTP, Cron, Kafka, NATS triggers • Simple HTTPRouting and Scheduled tasks	• Cold-start latency • Hard to debug across distributed pods
Elasticsearch	• Kibana UI, Beats and Logstash integration • Powerful aggregations and time-series rollups • Automatic sharding and high availability	• Slow parent-child / nested joins • Default 10 000-record retrieval limit

Table 3: Summary of Pros and Cons for Key Technologies

5.5 Challenges during this project

5.5.1 Data Acquisition and Volume Requirements

Scraping even 500,000 Reddit comments yielded only around 100 MB of raw text. Our analytical goals, however, demanded on the order of 100 GB of data to ensure statistical robustness across time periods and regions. Bridging this gap required careful planning to continuously expand our dataset over weeks or months.

Additionally, to respect platform rate limits, we could not pull all historical posts in one go. Instead, we relied on an automated, time-based batch process via Fission HTTP Triggers to periodically harvest the latest comments. This approach gradually accumulated sufficient volume but introduced latency in having a complete dataset for any given window.

5.5.2 API Limitations

The official Reddit API does not support arbitrary time-range queries by timestamp, forcing us to harvest chronologically and then filter locally. This made it impossible to jump directly to a specific date range, complicating efforts to fill gaps in older data. Because our scheduled scrapes only capture new posts, historical coverage depends on the consistency of our scraping jobs. Any downtime or errors in the pipeline risked permanent data gaps that could bias our election-period analyses.

Both Reddit and Mastodon impose strict limits on request rates. Exceeding these limits triggers temporary blocks, so our scraper had to implement exponential back-off and retry logic. Balancing aggressiveness (to gather data quickly) with politeness (to avoid bans) required extensive tuning.

5.5.3 Query Size Limitation in Elasticsearch

One significant challenge encountered during this project was the default limit of 10,000 documents imposed by Elasticsearch’s `search` API (`index.max_result_window = 10000`). This restriction meant that for large result sets such as those generated when querying popular keywords or long date ranges, only the first 10,000 documents would be returned, potentially truncating important data without warning.

To address this, the query process was designed to recursively divide the target date range into smaller subranges. For each range, we performed a count check and if the number of hits exceeded the threshold, the range was further split until all returned batches contained fewer than 10,000 documents. This method ensured complete data retrieval while avoiding the complexity and stateful management required by `scroll` contexts.

6 Team Contributions

Our team adopted a collaborative approach throughout the development of the project, intentionally avoiding a rigid assembly-line structure. Instead of dividing the project into isolated parts, we progressed through each phase together. From selecting our own scenarios to implementing data harvesting, Elasticsearch indexing, ReST API development, analysis, and the final presentation in

Jupyter Notebook, each team member engaged with the project in a comprehensive and integrative manner.

In the initial weeks, we met twice a week to coordinate our progress and discuss technical directions. As the project advanced, meeting frequency increased to ensure smooth integration and timely completion. Throughout, we communicated actively regarding our individual progress, shared insights, and resolved technical issues together. Collaboration was rooted in technical understanding, and by the end of the project, every team member had gained familiarity with all components of the system. This collective understanding enabled us to consolidate our work effectively for the final submission.

One of the challenges we encountered was the diversity of topics explored at the beginning, which led to a somewhat fragmented focus. This issue was addressed midway through the project when we combined and refined our topics to ensure consistency. Additionally, while our collaborative method may have resulted in a slower pace compared to a linear task distribution, it allowed all team members to remain on the same page and contributed to the robustness of our final product.

7 Conclusion and Future Work

In this project, we developed a scalable cloud-native analytics platform that integrates unstructured social media data with structured government datasets to explore real-world trends across Australian society. Hosted on the Melbourne Research Cloud, the system leverages Kubernetes for orchestration, Fission for event-driven ingestion, and Elasticsearch for scalable data storage and querying. This architecture enabled efficient ingestion and processing of heterogeneous data sources, supporting a broad range of analyses from urban traffic trends to sentiment around elections, inflation, and crime.

Our analysis yielded several key insights. We found a moderately negative relationship between housing sentiment and property prices post-2022, suggesting growing public frustration with affordability. Reddit voting trends were moderately correlated with official inflation data, affirming the platform’s potential as an early signal for public economic concerns. Lagged analysis also revealed a weak but notable temporal link between YouTube video popularity and future traffic congestion, indicating that online engagement may precede changes in mobility patterns.

Throughout this project, we encountered several limitations, including sparse data in earlier years, limitations of keyword-based sentiment detection, and the imprecision of public API search mechanisms. These challenges underline the need for more advanced NLP models and improved data retrieval techniques.

Future work should focus on improving sentiment detection using transformer-based NLP models capable of handling nuanced language features such as sarcasm and idioms. The system could be enhanced further by incorporating geospatial resolution to support localised analyses and by integrating forecasting and causal inference tools to evaluate the predictive utility of online sentiment. Expanding the frontend with real-time dashboards would improve accessibility for end users, while broadening data collection to include additional social media platforms would help mitigate bias and increase coverage.

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